

SkyHAD: Cooperative Aerial Asset Defense and Sixteen Native Task Formulations

Abstract—Cooperative Red defenders protect stationary assets against Blue opponents in a three-dimensional scene. Eight native model–action interfaces and two objectives define sixteen tasks. Particle, fixed-wing and quadrotor dynamics give different control burdens to acceleration, position and actuator commands. Survival protects every asset through a terminal team payoff; Damage minimizes cumulative raw asset damage. This formulation distinguishes local actor information, privileged critic features, complete transition state and external opponent policies, and states the assumptions for decentralized execution and centralized training.

Index Terms—Multi-agent reinforcement learning, cooperative defense, decentralized control, aerial simulation.

I. SCENARIO AND RESEARCH GOAL

Red agents coordinate interception and protection of stationary assets. Blue agents navigate under a specified external opposing policy; automatic Attack fire and other role interactions are resolved independently of navigation commands. The ultimate research goal is to learn effective cooperative Red defense with decentralized policies acting on local information. Training may use explicitly provided centralized features.

Three physical models are available: a six-value kinematic particle, a six-degree-of-freedom aerodynamic fixed wing, and a six-degree-of-freedom ideal quadrotor. Model type differs from the three episode-fixed functional roles: Attack, Scout and Disturb. Every role uses its selected model and navigation interface; functional decisions follow shared environment rules.

The particle supports acceleration and absolute-position actions. Each rigid model supports acceleration, direct actuator and absolute-position actions. These are eight native model–action combinations. Pairing each with Survival and Damage gives the sixteen themes indexed on page 4.

A. Task distinction

Survival uses default terminal Red payoff ± 10 with asset-loss priority. Damage accumulates raw asset losses while

finite asset health remains unchanged. Red extinction alone is not a native global terminal event. The sample horizon H limits collected trajectories; it does not replace the continuing objective by a finite-horizon game.

B. Scope and reading order

Pages 2–4 define shared dynamics, ordered events, information, tasks and learning. Each theme then provides one physical/action page and one game/learning page. Only particle themes admit planar compatibility; the native formulations are 3D. The document has 36 pages.

The implementation grounds this specification: particle position advances with old velocity; all living entities appear in native actor rows; automatic firing is separate from external Blue navigation; and fixing Blue is an explicit experiment choice. Physical fidelity and action abstraction are experimental choices, so the control burden differs across interfaces.

C. Units and model assumptions

Distances are metres, speeds m/s, rates rad/s and time seconds. λ scales scene geometry and role radii, never aircraft mass/inertia. $\text{sat}_L(z)$ limits Euclidean norm; clip limits components/scalars. Each physics page supplies its own ρ_0 , physical parameters, controller and action interface.

D. Implementation correspondence and assumptions

The current flat implementation grounds this specification: `config.py` fixes scales and limits; `initialization.py` defines reset sampling; `dynamics/` defines particle/rigid maps, controllers and physical constants; `world.py` and `simulation.py` resolve events and motion; `observations.py` packs information; `tasks.py` defines rewards and completion; `grouping/` defines the optional extension. `docs/MODELING.md` records the same modeling assumptions.

Physical parameters and source-derived attitude gains are attributed in the physics sections [1], [2], [3]. Interface guidance, action scalings, finite asset health, boundaries and task objectives are project choices. No wind, observation noise, actuator lag or unlisted vehicle drag is assumed. These sources establish model assumptions, not training results for the sixteen instances.

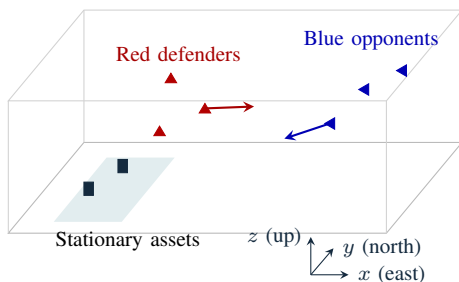


Fig. 1. Illustrative 3D asset-defense scene. Arrows show navigation directions; no sensing or firing range is depicted.

II. COMMON TRANSITION AND PHYSICAL STATE

A. Sets, symbols and complete state

Red defenders $\mathcal{R}$ protect stationary assets $\mathcal{P}$ against Blue $\mathcal{B}$. Set $\mathcal{U} = \mathcal{R} \cup \mathcal{B}$, $\mathcal{E} = \mathcal{U} \cup \mathcal{P}$, $N = |\mathcal{U}|$, $K = |\mathcal{P}|$, $M = N + K$. Role $r_i \in \{A, S, D\}$ (Attack, Scout, Disturb), side $c_i \in \{R, B\}$ and roster identity id_i are episode-fixed. Set subscripts select roles.

$$s_t = (\{x_i, h_i, r_i, c_i, \text{id}_i\}_{i \in \mathcal{U}}, \{p_k, h_k, D_k\}_{k \in \mathcal{P}}, t, \zeta_t). \quad (1)$$

Particle $x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$; rigid $x_i = (p_i, v_i, q_i, \omega_i) \in \mathbb{R}^{13}$, $\|q_i\| = 1$. Participation is $\ell_i = \mathbf{1}[h_i > 0]$. Reset UAV health is 1, target health $h_k^0 = 2$ by default; $v_k = 0$, $D_k = 0$ initially. D_k accumulates raw asset damage.

B. Six-degree-of-freedom dynamics

Use world ENU and body FLU axes. The scalar-first unit quaternion $q = (q_w, q_x, q_y, q_z) \sim -q$ maps body to world through $R(q) \in SO(3)$. The 13 stored values represent a six-DOF rigid body with a 12-dimensional state: six configuration and six velocity coordinates.

$$\begin{aligned} \dot{p} &= v, & \dot{v} &= R(q)F_B/m - g e_3, \\ \dot{q} &= \frac{1}{2}q \otimes (0, \omega), & \dot{\omega} &= J^{-1}(M_B - \omega \times J\omega). \end{aligned} \quad (2)$$

Here $g = 9.81 \text{ m/s}^2$, ω is body angular velocity (rad/s), and F_B excludes gravity; wind, sensor noise and actuator lag are absent.

$$\begin{aligned} R(q) &= I + 2q_w[q_v]_{\times} + 2[q_v]_{\times}^2, \\ q_v &= (q_x, q_y, q_z). \end{aligned} \quad (3)$$

Here $[z]_{\times} y = z \times y$.

C. Numerical transition and boundary

Write $f(x, u)$ for the above derivative, evaluated with normalized q . An external step is $\Delta = 1 \text{ s}$, $n = \lceil \Delta / .01 \rceil = 100$, $h = \Delta / n$. With stage feedback U ,

$$\begin{aligned} k_1 &= f(x, U(x)), \\ k_2 &= f(x + hk_1/2, U(x + hk_1/2)), \\ k_3 &= f(x + hk_2/2, U(x + hk_2/2)), \\ k_4 &= f(x + hk_3, U(x + hk_3)), \\ x^+ &= x + h(k_1 + 2k_2 + 2k_3 + k_4)/6, \\ q^+ &\leftarrow q^+ / \|q^+\|. \end{aligned} \quad (4)$$

Repeat n times with normalized stage quaternions. Hold actuators or intent references; recompute intent feedback at each stage. Controllers have no hidden integrators.

Clip endpoint $p'_a = \text{clip}(p_a^*, b_a^-, b_a^+)$. At $p_a^* > b_a^+$ set $v'_a = \min(v_a^*, 0)$; at $p_a^* < b_a^-$ set $v'_a = \max(v_a^*, 0)$; otherwise retain v_a^* . Clip collision-path substep positions too. Controller reference limits do not clamp physical airspeed or attitude.

D. Collision and automatic interaction

Predict living UAV paths: particles use $p_i + \tau \Delta v_i$, rigid models use synchronized RK4 endpoints clipped to $\mathcal{W}$. On each matched segment, let $r = p_i - p_j$, $d = (q_i - p_i) - (q_j - p_j)$:

$$\tau_* = \begin{cases} 0 & \|d\|^2 \leq 10^{-12}, \\ \text{clip}(-r^T d / \|d\|^2, 0, 1) & \text{otherwise,} \end{cases} \quad (5)$$

$$d_* = \|r + \tau_* d\|.$$

Collision sets both UAV HP to zero when $\min d_* \leq d_c$; targets are excluded. Pairs use i -major, $j < i$ order. Dead outer agents are skipped at entry and dead inner agents are skipped. An outer agent killed during its inner sweep continues that sweep.

After collision, surviving Attack agents automatically fire at step-start positions:

$$I_i = \begin{cases} \mathbf{1}[\min_{k \in \mathcal{P}: h_k > 0} \|p_i - p_k\| < r_f], & i \in \mathcal{B}, \\ \mathbf{1}[\min_{j \in \mathcal{B}_A: h_j > 0} \|p_i - p_j\| < r_f], & i \in \mathcal{R}. \end{cases} \quad (6)$$

An empty minimum is $+\infty$; dead sources have $I_i = 0$. A denotes Attack. Fire is a separate rule, not a navigation action. With unit attack intensity,

$$\eta(d) = \begin{cases} 1 & d < r_0, \\ (r_1 - d)/(r_1 - r_0) & r_0 \leq d < r_1, \\ 0 & d \geq r_1, \end{cases} \quad (7)$$

$$\begin{aligned} L_k &= \sum_{b \in \mathcal{B}_A: h_b > 0} I_b \eta(\|p_b - p_k\|), \\ L_i^{\text{atk}} &= \sum_{j \in \mathcal{U}_A: c_j \neq c_i, h_j > 0} I_j \eta(\|p_j - p_i\|). \end{aligned} \quad (8)$$

A source can affect multiple enemies, excluding friendly explosions. Sum losses on one post-collision, pre-motion snapshot before clipping HP.

E. Disturb and Scout

For source j , $d_{ji} = \|p_i - p_j\|$ and $\theta_{ji} = \angle(v_j, p_i - p_j)$; angle is zero if either norm is below 10^{-3} . Let

$$\begin{aligned} G_{ji} &= \mathbf{1}[10^{-3} < d_{ji} < 600\lambda, \theta_{ji} < \pi/6] \\ &\quad \times e^{-(d_{ji}/\lambda)^2/2} / \sqrt{2\pi}, \\ S_c(j) &= \sum_{i \in \mathcal{U}: c_i = c, h_i > 0} G_{ji}. \end{aligned} \quad (9)$$

Living Red Disturb uses $J_j = \mathbf{1}[S_R(j) < S_B(j)]$, Blue uses $J_j = \mathbf{1}[S_R(j) \geq S_B(j)]$. Dead sources contribute zero.

$$\begin{aligned} L_i^{\text{dis}} &= .1 \sum_{j \in \mathcal{U}_D: h_j > 0} J_j G_{ji}, \\ \tilde{h}_i &= \max(0, h_i - L_i^{\text{atk}} - L_i^{\text{dis}}). \end{aligned} \quad (10)$$

Set $h'_i = 0$ if $r_i = A$, $I_i = 1$, otherwise $h'_i = \tilde{h}_i$. Disturb damages either side, not assets. Scouts use $d < 2000\lambda$, $\theta < \pi/3$; this sector does not restrict native actor rows. Collision victims do not move. Combat victims complete their motion; inactive velocity is zeroed next step. The native order is collision, role decision, losses and source consumption, motion, asset/task update, cycle increment, then reward, completion/truncation and observation. Snapshot pre-step participation; retain commands, flags and raw damage for replay.

III. COMMON INFORMATION AND DEFENDER TASK

A. Observation and information structure

For each living entity $j \neq i$, the actor row is

$$\begin{aligned} e_{ij} &= [(p_j - p_i)/L, (v_j - v_i)/(2V_{\max}), \\ &\quad h_j, 1, \mathbf{1}_{c_j=R}, \mathbf{1}_{c_j=B}, \mathbf{1}_{j \in \mathcal{P}}], \\ L &= \sqrt{\sum_a (b_a^+ - b_a^-)^2}. \end{aligned} \quad (11)$$

Here $V_{\max}$ is the preset maximum speed, not the reference speed. Rows have 11 values, in fixed roster order: Red Scout/Disturb/Attack, Blue Scout/Disturb/Attack, then targets, excluding self. Dead rows are zero. Every living entity is included; Attack, Scout and Disturb share this packing. Side flags do not encode role. Own health is absent from the self block.

Interface-specific self blocks and dimensions are stated in each theme. With deterministic encoding, $Z_i(o \mid s) = \delta_{\text{encode}_i(s)}(o)$. Other UAV attitudes/rates, opponent memory and time are omitted; hence local information is partial despite noiseless sensing.

Privileged $g(s) = \text{state}()$ packs 11 values per entity,

$$[(p - b^-)/(b^+ - b^-), v/(2V_{\max}), h, \ell, R, B, P], \quad (12)$$

with dead blocks zero. Rigid models append 13 raw state values per UAV (dead zero), then $[t, \sum_k D_k, \mathbf{1}_{\text{Damage}}]$: size $11M + 13N + 3$. Particle size is $11M$, omitting time and damage. g is a CTDE feature, not the complete Markov state.

B. Fixed-opponent defender Dec-POMDP

For specified frozen external navigation policy π_B , including its memory transition, define

$$\begin{aligned} \mathcal{M}_R &= (\mathcal{R}, \mathcal{S}, \{\mathcal{A}_i\}, T^{\pi_B}, \{\mathcal{O}_i\}, Z, R, \gamma, \rho_0, H), \\ T^{\pi_B}(s' \mid s, a_R) &= \int \delta_{F_C(s, a_R, a_B)}(s') \pi_B(da_B \mid o_B, \zeta_t). \end{aligned} \quad (13)$$

F_C includes the ordered events, model-specific controlled motion and any memory update. Native execution accepts both teams' navigation actions. Automatic fire is an environment rule and does not supply Blue navigation. A rule-based or learned frozen Blue policy is an explicit experiment choice; with both teams learning the system is a partially observable stochastic game (POSG). The reset law ρ_0 is defined by the model physics page.

C. Configuration and replay state

Configuration C fixes roster, bounds, controls and task. ζ_t contains persistent assignments and opponent memory, if used. Reactive controllers have no hidden integrators. Last command, functional bits, pre-step positions and step damage are overwritten transition records; retain them for exact reward/event replay. RNG state is required for exact replay, not actor input.

D. Interpretation of partial information

Actor inputs omit other UAV attitudes/rates, policy memory and time. Fixed roster positions and side flags do not encode role directly. Full entity inclusion therefore does not guarantee Markov observations. A recurrent policy's local history can retain its own inferred information; that policy memory also affects exact centralized value targets. Privileged $g(s)$ is an encoding for CTDE, not a complete simulation snapshot or proof of the Markov property.

E. Survival payoff and completion

Assets obey $h'_k = \max(0, h_k - L_k)$, $D'_k = D_k + L_k$.

$$z(s') = \begin{cases} -1 & \exists k : h'_k < 10^{-3}, \\ +1 & \forall b \in \mathcal{B}_A : h'_b < 10^{-3}, \\ 0 & \text{otherwise.} \end{cases} \quad (14)$$

Target loss has priority. Natural completion is $d = \mathbf{1}[z \neq 0]$; default Red reward $R_t = 10z$, Blue $-R_t$. Red elimination alone does not end the world. Individual $h'_i \leq 0$ retires a Survival actor after its final transition. Horizon H truncates without a win bonus; use global bootstrap $b_t = 1 - d$, including individual death and truncation. Actor action-loss masks use pre-step participation. Explicit reward weights replace the default payoff as defined on page 4.

F. Damage payoff and completion

Assets keep finite reset health $h'_k = h_k^0$. Count raw contributions before HP clipping, including simultaneous overkill:

$$D'_k = D_k + L_k, \quad d_t = \sum_k L_k, \quad R_t = -d_t. \quad (15)$$

Blue reward is d_t ; there is no team-size, asset-count or HP normalization and no terminal Survival bonus.

$$d = \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0]. \quad (16)$$

All agent slots remain until global completion/truncation. Dead slots have zero observations, ignored actions and still receive full team R_t , including later damage after Red extinction. Do not sum duplicate teammate rewards. Undiscounted returns are $(-\sum_k D_k, +\sum_k D_k)$. H only truncates; global bootstrap $b_t = 1 - d$ remains one at truncation. Actor action-loss masks use pre-step participation.

G. Three distinct masks

Entity validity masks identify living packed rows. Pre-step participation masks decide which actor actions contribute to the action loss, including a final lethal action. The global bootstrap mask suppresses successor value only at natural world completion. Actor retirement, entity death, sample truncation and global completion are separate events. When native completion coincides with H , natural completion takes priority.

IV. COMMON LEARNING OBJECTIVE AND IMPLEMENTATION INDEX

A. MARL / CTDE objective and value history

$$\begin{aligned}\tau_i &= (o_{i,0}, a_{i,0}, \dots, o_{i,t}), \\ \pi_R(a_R | \tau_R) &= \prod_{i \in \mathcal{R}} \pi_i(a_i | \tau_i), \\ \pi_R^* &\in \arg \max_{\pi_R} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t R_t \right].\end{aligned}\quad (17)$$

Expectation is over ρ_0, T^{π_B}, π_R . Here T is natural completion (possibly infinite); H limits sampled rollouts. A truncated return is partial, so its successor value is retained. For $\gamma = 1$, the native task is terminal Survival payoff or negative total asset damage; $\gamma < 1$ selects a discounted task. Actors use local histories only. Let ξ_t augment $g(s_t)$ with task state and all relevant policy history/memory, including recurrent Red actor memory and frozen Blue memory. Exact value targets condition on ξ_t ; $g(s)$ alone need not suffice:

$$y_t = R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \quad (18)$$

Here Q^- is a target critic and successor actions follow the target actor policies. Use global completion, not individual death. A recurrent critic or explicit memory/state augmentation supplies the required conditioning; privileged packed features do not automatically define a Markov critic state. This defines an objective; it does not prescribe an implemented optimizer.

B. Optional grouping / SMDP extension

Grouping adds assignments, reserves, decision clocks and policy memory to state, and currently executes acceleration commands. It does not add native themes. Macro duration d ends at a command opportunity, casualty or completion. With no living Red, the call continues Blue scheduling to completion/horizon and records `no_red_continuation=True`.

$$\bar{R}_t = \sum_{j=0}^{d-1} \gamma^j R_{t+j}, \quad y_t = \bar{R}_t + \gamma^d b_t V(\xi_{t+d}). \quad (19)$$

Current grouping Damage returns the *undiscounted* sum, directly matching $\gamma = 1$. A γ^d bootstrap alone does not discount within-macro rewards. Grouping Survival uses success 0/1 and a configured horizon result (`red_win`, `draw`, `blue_win`); native Survival instead uses ± 10 and sampling truncation. Grouping Damage retains sampling truncation.

Built-in fixed Blue opponents are reactive, balanced and concentrated with a rush executor. Balanced uses target quotas; concentrated selects minimum mean-travel target. Reactive target probability is proportional to $e^{-2 \text{coverage} - .25 \text{travel}}$, with coverage $\sum_{i \in \mathcal{R}: h_i > 0} h_i e^{-d_{ik}/(850\lambda)}$ and travel equal to mean Blue distance divided by 5000λ . Native seedless resets advance the local RNG; grouping seedless reset reproduces its saved seeded initialization.

TABLE I
EIGHT NATIVE MODEL-ACTION COMBINATIONS: THEME IDS AND GENERATED PHYSICS-GAME PAGE RANGES.

Model	Action	Survival	Damage
Particle	Acceleration	1: 5–6	2: 7–8
Particle	Position	3: 9–10	4: 11–12
Fixed wing	Acceleration	5: 13–14	6: 15–16
Fixed wing	Actuator	7: 17–18	8: 19–20
Fixed wing	Position	9: 21–22	10: 23–24
Quadrotor	Acceleration	11: 25–26	12: 27–28
Quadrotor	Actuator	13: 29–30	14: 31–32
Quadrotor	Position	15: 33–34	16: 35–36

C. Optional native Survival rewards

Each formulation uses its native default payoff. An explicit four-vector w replaces Survival scalar reward by $w^T \ell_i$, where $\ell_i = (r_i^{\text{hit}}, r_i^{\text{target}}, r_i^{\text{enemy}}, 10\epsilon_i z)$, where $\epsilon_i = +1$ for Red and -1 for Blue. All geometry below is post-step. For a living Attack agent,

$$\begin{aligned}r_i^{\text{hit}} &= \sum_{j \in \mathcal{U}: c_j \neq c_i, h_j > 0} \max(0, 1 - d_{ij}/r_1), \\ r_i^{\text{target}} &= - \min_{k: h_k > 0} d_{ik}/L, \\ r_i^{\text{enemy}} &= \begin{cases} - \min_{j \in \mathcal{B}: h_j > 0} d_{ij}/L & i \in \mathcal{R}, \\ \min_{j \in \mathcal{R}: h_j > 0} d_{ij}/L - 1 & i \in \mathcal{B}. \end{cases}\end{aligned}\quad (20)$$

An empty comparison set gives zero, including the Blue enemy slot; dead agents have zero first three slots. “Hit” is a geometric proxy, not actual fire damage. Scout/Disturb keep their role scalar in slot 1 and zero slots 2–3. Let δ_i be nearest XY-wall distance and $B_i = .05e^{-\delta_i/\lambda} \mathbf{1}[\delta_i < 100\lambda]$. Scout slot 1 is $c_s |\text{sector}_i| - B_i$; Disturb slot 1 is $\mathbf{1}[J_i, h_i > 0](.1c_d \sum_{j \in \mathcal{U}} \text{sign}_{ij} G_{ij} - B_i)$, with enemy sign $+1$, friendly sign -1 . Defaults $c_s = c_d = 0$. The episode slot already contains the team payoff; adding it again doubles the terminal reward. Damage rejects these weights.

REFERENCES

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- [3] M. Faessler, A. Franchi, and D. Scaramuzza, “Differential Flatness of Quadrotor Dynamics Subject to Rotor Drag for Accurate Tracking of High-Speed Trajectories,” RAL, 2018. [Online]. Available: https://rpg.ifi.uzh.ch/docs/RAL18_Faessler.pdf. The ideal project model omits those drag terms.

V. THEME 1 / PARTICLE / ACCELERATION / SURVIVAL

Physical state, navigation action and discrete transition.

A. Physical state for this interface

$x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$ is a six-value kinematic state; no quaternion, inertia, wing or rotor is attached to a particle. The complete state, health, role, side and replay records follow page 2.

B. Implemented kinematic map

For a living, noncolliding particle, $\Delta = 1$ s,

$$\begin{aligned} p_i^* &= p_i + \Delta v_i, & p_i' &= \Pi_{\mathcal{W}}(p_i^*), \\ w_i &= v_i + \Delta \alpha_i, & \tilde{v}_i &= \mathcal{S}_{35,120}(w_i; v_i). \end{aligned} \quad (21)$$

For $\|w\| \geq 10^{-3}$,

$$\mathcal{S}_{m,V}(w; v) = \frac{w}{\|w\|} \text{clip}(\|w\|, m, V). \quad (22)$$

For $\|w\| < 10^{-3}$, use $mv/\|v\|$ when $\|v\| \geq 10^{-3}$ and me_1 otherwise. The cancellation fallback retains heading. After speed normalization, an outward component is set to zero on each clipped axis: upper wall $v_a > 0$ or lower wall $v_a < 0$.

The final 3D turn map limits the angle between old and candidate velocities to $\theta_{\max} = \pi$ per step, so the default map is identity. For a smaller cap, rotate the old heading toward the candidate about $v_i \times \tilde{v}_i$, preserving candidate speed. A deterministic orthogonal axis resolves the antiparallel case; near-zero products $\|v\|_i \|\tilde{v}_i\| \leq 10^{-3}$ have angle zero.

Position uses old velocity, with no $\frac{1}{2}\alpha\Delta^2$ term. Boundary projection can reduce the final speed below 35 m/s. Agents killed by collision do not move; agents killed by combat complete this step's motion, and their velocity is zeroed on the next inactive update.

C. Interpretation

This is a kinematic benchmark. Its action controls future velocity and thus subsequent interception geometry, rather than instantaneous translation. Flight decisions and automatic functional decisions are distinct.

D. Acceleration intent

Each living agent chooses either $a_i \in [-1, 1]^3$ or an index into

$$\mathcal{V} = \{0\} \cup \{z/\|z\| : z \in \{-1, 0, 1\}^3 \setminus \{0\}\}, \quad (23)$$

giving 27 discrete primitives, zero first. The physical intent is $\alpha_i = 40a_i$ m/s². Continuous inputs have component bounds, so diagonal intents can have norm $\sqrt{3}40$; discrete nonzero intents have norm 40. 40 is a control scale, not a clamp on actual rigid-body acceleration. The chosen vector directly enters the particle map as $\alpha_i = 40a_i$. No inner rigid-body controller or actuator state is introduced. Zero input preserves unconstrained velocity; it does not command a stop.

E. Episode geometry and initial law

The default scale is $\lambda = 1$, with

$$\mathcal{W} = \lambda([-2500, 2500]^2 \times [0, 2500]) \text{ m}. \quad (24)$$

Assets are sampled independently from $\lambda([-2300, -1900] \times [-1200, 1200] \times [500, 1500])$, or supplied as a fixed layout. Red chooses a target uniformly and samples an area-uniform XY annulus: $\rho^2 \sim U((800\lambda)^2, (2200\lambda)^2)$, $\vartheta \sim U(0, 2\pi)$; XY is clipped to $\mathcal{W}$. Blue samples $x \sim U(0, 2500\lambda)$ and $y \sim U(-2500\lambda, 2500\lambda)$. UAV altitude is sampled in $[200, 1500]\lambda$; target altitude is separate.

Random velocity has a Gaussian unit direction and speed $\|(u_1, u_2, u_3)\|/\sqrt{3}$, where $u_a \sim U(35, 120)$. Initial positions and velocities are generated by the environment-local RNG. The default collision distance is $d_c = 20\lambda = 20$ m. The full/zero-damage radii are $(r_0, r_1) = (300, 600)\lambda = (300, 600)$ m. The default firing threshold is $r_f = r_0$; configuration may select $0 < r_f \leq r_0$.

F. 2D compatibility

Only particle themes support $d = 2$. The three coordinate slots remain, $p_z = 100\lambda$, $v_z = \alpha_z = 0$. Acceleration uses a 2-vector or nine planar primitives; position remains a 3-vector with prescribed altitude. Radii become $(200, 400)\lambda$. Planar resets reject too-close openings, and the heading cap is applied on the circle. These are compatibility settings of this theme, not additional themes.

G. Task dependence

Navigation and observations are identical between Survival and Damage. Asset health evolution, completion, and defender reward differ as specified on the next page. Neither task changes the particle into a rigid-body vehicle.

VI. THEME 1 / PARTICLE / ACCELERATION / SURVIVAL

Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 5, combined with the shared event rules on page 2, defines F_C . The configuration selects **Particle / Acceleration** and **Survival**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Particle event and scale instance

Use the six-value kinematic state, $\lambda = 1$, $d_c = 20$ m and $(r_0, r_1) = (300, 600)$ m. Actor normalization uses $V_{\max} = 120$ m/s. Collision prediction uses old-velocity line segments, and position advances before the chosen acceleration affects future velocity. The 35–120 m/s speed normalization and turn map belong to this physical interface; clipping at a wall can reduce speed below 35 m/s. The acceleration action has three continuous coordinates or 27 unit primitives. The actor lacks a self-state block, so it must infer its own motion from its available history when needed; relative entity rows alone do not supply absolute velocity or absolute position. Only this model admits the planar compatibility setting on its physics page. All equations and dimensions here describe the native 3D instance.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor input is only an $(M-1) \times 11$ entity table ($V_{\max} = 120$); it has no own-state block. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\dim o_i = 11(M-1), \quad \dim g = 11M. \quad (25)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Survival reward and completion instance

Apply the Survival rule on page 3: asset HP decreases by raw L_k , and target loss has priority over Blue Attack elimination.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\text{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \quad (26)$$

Blue receives $-R_t$. Red elimination alone does not end the world. A nonpositive-health actor retires after its lethal final transition, while the team return and successor value continue until global natural completion. Explicit optional reward weights replace this native scalar by the formulation on page 4; they do not add a second terminal bonus.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i \mid \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 5. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (27)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

VII. THEME 2 / PARTICLE / ACCELERATION / DAMAGE

Physical state, navigation action and discrete transition.

A. Physical state for this interface

$x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$ is a six-value kinematic state; no quaternion, inertia, wing or rotor is attached to a particle. The complete state, health, role, side and replay records follow page 2.

B. Implemented kinematic map

For a living, noncolliding particle, $\Delta = 1$ s,

$$\begin{aligned} p_i^* &= p_i + \Delta v_i, & p_i' &= \Pi_{\mathcal{W}}(p_i^*), \\ w_i &= v_i + \Delta \alpha_i, & \tilde{v}_i &= \mathcal{S}_{35,120}(w_i; v_i). \end{aligned} \quad (28)$$

For $\|w\| \geq 10^{-3}$,

$$\mathcal{S}_{m,V}(w; v) = \frac{w}{\|w\|} \text{clip}(\|w\|, m, V). \quad (29)$$

For $\|w\| < 10^{-3}$, use $mv/\|v\|$ when $\|v\| \geq 10^{-3}$ and me_1 otherwise. The cancellation fallback retains heading. After speed normalization, an outward component is set to zero on each clipped axis: upper wall $v_a > 0$ or lower wall $v_a < 0$.

The final 3D turn map limits the angle between old and candidate velocities to $\theta_{\max} = \pi$ per step, so the default map is identity. For a smaller cap, rotate the old heading toward the candidate about $v_i \times \tilde{v}_i$, preserving candidate speed. A deterministic orthogonal axis resolves the antiparallel case; near-zero products $\|v\|_i \|\tilde{v}_i\| \leq 10^{-3}$ have angle zero.

Position uses old velocity, with no $\frac{1}{2}\alpha\Delta^2$ term. Boundary projection can reduce the final speed below 35 m/s. Agents killed by collision do not move; agents killed by combat complete this step's motion, and their velocity is zeroed on the next inactive update.

C. Interpretation

This is a kinematic benchmark. Its action controls future velocity and thus subsequent interception geometry, rather than instantaneous translation. Flight decisions and automatic functional decisions are distinct.

D. Acceleration intent

Each living agent chooses either $a_i \in [-1, 1]^3$ or an index into

$$\mathcal{V} = \{0\} \cup \{z/\|z\| : z \in \{-1, 0, 1\}^3 \setminus \{0\}\}, \quad (30)$$

giving 27 discrete primitives, zero first. The physical intent is $\alpha_i = 40a_i$ m/s². Continuous inputs have component bounds, so diagonal intents can have norm $\sqrt{3}40$; discrete nonzero intents have norm 40. 40 is a control scale, not a clamp on actual rigid-body acceleration. The chosen vector directly enters the particle map as $\alpha_i = 40a_i$. No inner rigid-body controller or actuator state is introduced. Zero input preserves unconstrained velocity; it does not command a stop.

E. Episode geometry and initial law

The default scale is $\lambda = 1$, with

$$\mathcal{W} = \lambda([-2500, 2500]^2 \times [0, 2500]) \text{ m}. \quad (31)$$

Assets are sampled independently from $\lambda([-2300, -1900] \times [-1200, 1200] \times [500, 1500])$, or supplied as a fixed layout. Red chooses a target uniformly and samples an area-uniform XY annulus: $\rho^2 \sim U((800\lambda)^2, (2200\lambda)^2)$, $\vartheta \sim U(0, 2\pi)$; XY is clipped to $\mathcal{W}$. Blue samples $x \sim U(0, 2500\lambda)$ and $y \sim U(-2500\lambda, 2500\lambda)$. UAV altitude is sampled in $[200, 1500]\lambda$; target altitude is separate.

Random velocity has a Gaussian unit direction and speed $\|(u_1, u_2, u_3)\|/\sqrt{3}$, where $u_a \sim U(35, 120)$. Initial positions and velocities are generated by the environment-local RNG. The default collision distance is $d_c = 20\lambda = 20$ m. The full/zero-damage radii are $(r_0, r_1) = (300, 600)\lambda = (300, 600)$ m. The default firing threshold is $r_f = r_0$; configuration may select $0 < r_f \leq r_0$.

F. 2D compatibility

Only particle themes support $d = 2$. The three coordinate slots remain, $p_z = 100\lambda$, $v_z = \alpha_z = 0$. Acceleration uses a 2-vector or nine planar primitives; position remains a 3-vector with prescribed altitude. Radii become $(200, 400)\lambda$. Planar resets reject too-close openings, and the heading cap is applied on the circle. These are compatibility settings of this theme, not additional themes.

G. Task dependence

Navigation and observations are identical between Survival and Damage. Asset health evolution, completion, and defender reward differ as specified on the next page. Neither task changes the particle into a rigid-body vehicle.

VIII. THEME 2 / PARTICLE / ACCELERATION / DAMAGE

Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 7, combined with the shared event rules on page 2, defines F_C . The configuration selects **Particle / Acceleration** and **Damage**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Particle event and scale instance

Use the six-value kinematic state, $\lambda = 1$, $d_c = 20$ m and $(r_0, r_1) = (300, 600)$ m. Actor normalization uses $V_{\max} = 120$ m/s. Collision prediction uses old-velocity line segments, and position advances before the chosen acceleration affects future velocity. The 35–120 m/s speed normalization and turn map belong to this physical interface; clipping at a wall can reduce speed below 35 m/s. The acceleration action has three continuous coordinates or 27 unit primitives. The actor lacks a self-state block, so it must infer its own motion from its available history when needed; relative entity rows alone do not supply absolute velocity or absolute position. Only this model admits the planar compatibility setting on its physics page. All equations and dimensions here describe the native 3D instance.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor input is only an $(M - 1) \times 11$ entity table ($V_{\max} = 120$); it has no own-state block. In Damage, an already-dead observer has a zero complete table. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\dim o_i = 11(M - 1), \quad \dim g = 11M. \quad (32)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = 1[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely

from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Damage reward and completion instance

Apply the Damage rule on page 3: finite asset HP stays at reset health; raw simultaneous damage is counted before clipping or overkill normalization.

$$\begin{aligned} h'_k &= h_k^0, & D'_k &= D_k + L_k, \\ R_t &= -\sum_k L_k, & d_t^{\text{term}} &= 1[\forall b \in \mathcal{B}_A : h'_b \leq 0]. \end{aligned} \quad (33)$$

Blue receives $-R_t$. Every team slot receives the full shared reward, including dead Red slots and later damage after Red extinction; summing duplicate teammate rewards changes the objective. Dead slots keep zero observations and ignored actions until global completion/truncation. At $\gamma = 1$, the Red return is $-\sum_k D_k$, with no Survival terminal bonus or optional Survival weights.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i | \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 7. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (34)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

IX. THEME 3 / PARTICLE / POSITION / SURVIVAL

Physical state, navigation action and discrete transition.

A. Physical state for this interface

$x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$ is a six-value kinematic state; no quaternion, inertia, wing or rotor is attached to a particle. The complete state, health, role, side and replay records follow page 2.

B. Implemented kinematic map

For a living, noncolliding particle, $\Delta = 1$ s,

$$\begin{aligned} p_i^* &= p_i + \Delta v_i, & p_i' &= \Pi_{\mathcal{W}}(p_i^*), \\ w_i &= v_i + \Delta \alpha_i, & \tilde{v}_i &= \mathcal{S}_{35,120}(w_i; v_i). \end{aligned} \quad (35)$$

For $\|w\| \geq 10^{-3}$,

$$\mathcal{S}_{m,V}(w; v) = \frac{w}{\|w\|} \text{clip}(\|w\|, m, V). \quad (36)$$

For $\|w\| < 10^{-3}$, use $mv/\|v\|$ when $\|v\| \geq 10^{-3}$ and me_1 otherwise. The cancellation fallback retains heading. After speed normalization, an outward component is set to zero on each clipped axis: upper wall $v_a > 0$ or lower wall $v_a < 0$.

The final 3D turn map limits the angle between old and candidate velocities to $\theta_{\max} = \pi$ per step, so the default map is identity. For a smaller cap, rotate the old heading toward the candidate about $v_i \times \tilde{v}_i$, preserving candidate speed. A deterministic orthogonal axis resolves the antiparallel case; near-zero products $\|v\|_i \|\tilde{v}_i\| \leq 10^{-3}$ have angle zero.

Position uses old velocity, with no $\frac{1}{2}\alpha\Delta^2$ term. Boundary projection can reduce the final speed below 35 m/s. Agents killed by collision do not move; agents killed by combat complete this step's motion, and their velocity is zeroed on the next inactive update.

C. Interpretation

This is a kinematic benchmark. Its action controls future velocity and thus subsequent interception geometry, rather than instantaneous translation. Flight decisions and automatic functional decisions are distinct.

D. Absolute position intent

Each agent selects $a_i = p_i^{\text{des}} \in \mathcal{W}$ in metres. The stateless outer loop computes

$$\begin{aligned} d_i &= p_i^{\text{des}} - p_i, \\ v_i^{\text{des}} &= d_i \min\{.5, 120/\max(\|d_i\|, 10^{-12})\}, \\ \alpha_i &= \text{sat}_{40}(2(v_i^{\text{des}} - v_i)), \\ \text{sat}_L(z) &= z \min\{1, L/\|z\|\}, \quad \text{sat}_L(0) = 0. \end{aligned} \quad (37)$$

The command is an absolute point, not a displacement or teleport. The velocity reference can approach zero, but the particle speed map enforces its nonzero free-flight minimum; this interface does not promise arrival with rest. There are three continuous action coordinates and no position-action quantization.

E. Episode geometry and initial law

The default scale is $\lambda = 1$, with

$$\mathcal{W} = \lambda([-2500, 2500]^2 \times [0, 2500]) \text{ m}. \quad (38)$$

Assets are sampled independently from $\lambda([-2300, -1900] \times [-1200, 1200] \times [500, 1500])$, or supplied as a fixed layout. Red chooses a target uniformly and samples an area-uniform XY annulus: $\rho^2 \sim U((800\lambda)^2, (2200\lambda)^2)$, $\vartheta \sim U(0, 2\pi)$; XY is clipped to $\mathcal{W}$. Blue samples $x \sim U(0, 2500\lambda)$ and $y \sim U(-2500\lambda, 2500\lambda)$. UAV altitude is sampled in $[200, 1500]\lambda$; target altitude is separate.

Random velocity has a Gaussian unit direction and speed $\|(u_1, u_2, u_3)\|/\sqrt{3}$, where $u_a \sim U(35, 120)$. Initial positions and velocities are generated by the environment-local RNG. The default collision distance is $d_c = 20\lambda = 20$ m. The full/zero-damage radii are $(r_0, r_1) = (300, 600)\lambda = (300, 600)$ m. The default firing threshold is $r_f = r_0$; configuration may select $0 < r_f \leq r_0$.

F. 2D compatibility

Only particle themes support $d = 2$. The three coordinate slots remain, $p_z = 100\lambda$, $v_z = \alpha_z = 0$. Acceleration uses a 2-vector or nine planar primitives; position remains a 3-vector with prescribed altitude. Radii become $(200, 400)\lambda$. Planar resets reject too-close openings, and the heading cap is applied on the circle. These are compatibility settings of this theme, not additional themes.

G. Task dependence

Navigation and observations are identical between Survival and Damage. Asset health evolution, completion, and defender reward differ as specified on the next page. Neither task changes the particle into a rigid-body vehicle.

X. THEME 3 / PARTICLE / POSITION / SURVIVAL

Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 9, combined with the shared event rules on page 2, defines F_C . The configuration selects **Particle / Position** and **Survival**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Particle event and scale instance

Use the six-value kinematic state, $\lambda = 1$, $d_c = 20$ m and $(r_0, r_1) = (300, 600)$ m. Actor normalization uses $V_{\max} = 120$ m/s. Collision prediction uses old-velocity line segments, and position advances before the chosen acceleration affects future velocity. The 35–120 m/s speed normalization and turn map belong to this physical interface; clipping at a wall can reduce speed below 35 m/s. The absolute-position action has three continuous coordinates in $\mathcal{W}$. The outer loop supplies a norm-limited acceleration; the six raw self values provide the position and velocity needed for this reference interface. A position intent does not override collision or minimum free-flight speed. Only this model admits the planar compatibility setting on its physics page. All equations and dimensions here describe the native 3D instance.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M - 1) \times 11$ entity table ($V_{\max} = 120$) and the unnormalized six-value self block $[p_i, v_i]$. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\dim o_i = 11(M - 1) + 6, \quad \dim g = 11M. \quad (39)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Survival reward and completion instance

Apply the Survival rule on page 3: asset HP decreases by raw L_k , and target loss has priority over Blue Attack elimination.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\text{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \quad (40)$$

Blue receives $-R_t$. Red elimination alone does not end the world. A nonpositive-health actor retires after its lethal final transition, while the team return and successor value continue until global natural completion. Explicit optional reward weights replace this native scalar by the formulation on page 4; they do not add a second terminal bonus.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i | \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 9. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (41)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XI. THEME 4 / PARTICLE / POSITION / DAMAGE

Physical state, navigation action and discrete transition.

A. Physical state for this interface

$x_i = (p_i, v_i) \in \mathcal{W} \times \mathbb{R}^3$ is a six-value kinematic state; no quaternion, inertia, wing or rotor is attached to a particle. The complete state, health, role, side and replay records follow page 2.

B. Implemented kinematic map

For a living, noncolliding particle, $\Delta = 1$ s,

$$\begin{aligned} p_i^* &= p_i + \Delta v_i, & p_i' &= \Pi_{\mathcal{W}}(p_i^*), \\ w_i &= v_i + \Delta \alpha_i, & \tilde{v}_i &= \mathcal{S}_{35,120}(w_i; v_i). \end{aligned} \quad (42)$$

For $\|w\| \geq 10^{-3}$,

$$\mathcal{S}_{m,V}(w; v) = \frac{w}{\|w\|} \text{clip}(\|w\|, m, V). \quad (43)$$

For $\|w\| < 10^{-3}$, use $mv/\|v\|$ when $\|v\| \geq 10^{-3}$ and me_1 otherwise. The cancellation fallback retains heading. After speed normalization, an outward component is set to zero on each clipped axis: upper wall $v_a > 0$ or lower wall $v_a < 0$.

The final 3D turn map limits the angle between old and candidate velocities to $\theta_{\max} = \pi$ per step, so the default map is identity. For a smaller cap, rotate the old heading toward the candidate about $v_i \times \tilde{v}_i$, preserving candidate speed. A deterministic orthogonal axis resolves the antiparallel case; near-zero products $\|v\|_i \|\tilde{v}_i\| \leq 10^{-3}$ have angle zero.

Position uses old velocity, with no $\frac{1}{2}\alpha\Delta^2$ term. Boundary projection can reduce the final speed below 35 m/s. Agents killed by collision do not move; agents killed by combat complete this step's motion, and their velocity is zeroed on the next inactive update.

C. Interpretation

This is a kinematic benchmark. Its action controls future velocity and thus subsequent interception geometry, rather than instantaneous translation. Flight decisions and automatic functional decisions are distinct.

D. Absolute position intent

Each agent selects $a_i = p_i^{\text{des}} \in \mathcal{W}$ in metres. The stateless outer loop computes

$$\begin{aligned} d_i &= p_i^{\text{des}} - p_i, \\ v_i^{\text{des}} &= d_i \min\{.5, 120/\max(\|d_i\|, 10^{-12})\}, \\ \alpha_i &= \text{sat}_{40}(2(v_i^{\text{des}} - v_i)), \\ \text{sat}_L(z) &= z \min\{1, L/\|z\|\}, \quad \text{sat}_L(0) = 0. \end{aligned} \quad (44)$$

The command is an absolute point, not a displacement or teleport. The velocity reference can approach zero, but the particle speed map enforces its nonzero free-flight minimum; this interface does not promise arrival with rest. There are three continuous action coordinates and no position-action quantization.

E. Episode geometry and initial law

The default scale is $\lambda = 1$, with

$$\mathcal{W} = \lambda([-2500, 2500]^2 \times [0, 2500]) \text{ m}. \quad (45)$$

Assets are sampled independently from $\lambda([-2300, -1900] \times [-1200, 1200] \times [500, 1500])$, or supplied as a fixed layout. Red chooses a target uniformly and samples an area-uniform XY annulus: $\rho^2 \sim U((800\lambda)^2, (2200\lambda)^2)$, $\vartheta \sim U(0, 2\pi)$; XY is clipped to $\mathcal{W}$. Blue samples $x \sim U(0, 2500\lambda)$ and $y \sim U(-2500\lambda, 2500\lambda)$. UAV altitude is sampled in $[200, 1500]\lambda$; target altitude is separate.

Random velocity has a Gaussian unit direction and speed $\|(u_1, u_2, u_3)\|/\sqrt{3}$, where $u_a \sim U(35, 120)$. Initial positions and velocities are generated by the environment-local RNG. The default collision distance is $d_c = 20\lambda = 20$ m. The full/zero-damage radii are $(r_0, r_1) = (300, 600)\lambda = (300, 600)$ m. The default firing threshold is $r_f = r_0$; configuration may select $0 < r_f \leq r_0$.

F. 2D compatibility

Only particle themes support $d = 2$. The three coordinate slots remain, $p_z = 100\lambda$, $v_z = \alpha_z = 0$. Acceleration uses a 2-vector or nine planar primitives; position remains a 3-vector with prescribed altitude. Radii become $(200, 400)\lambda$. Planar resets reject too-close openings, and the heading cap is applied on the circle. These are compatibility settings of this theme, not additional themes.

G. Task dependence

Navigation and observations are identical between Survival and Damage. Asset health evolution, completion, and defender reward differ as specified on the next page. Neither task changes the particle into a rigid-body vehicle.

XII. THEME 4 / PARTICLE / POSITION / DAMAGE

Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 11, combined with the shared event rules on page 2, defines F_C . The configuration selects **Particle / Position** and **Damage**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Particle event and scale instance

Use the six-value kinematic state, $\lambda = 1$, $d_c = 20$ m and $(r_0, r_1) = (300, 600)$ m. Actor normalization uses $V_{\max} = 120$ m/s. Collision prediction uses old-velocity line segments, and position advances before the chosen acceleration affects future velocity. The 35–120 m/s speed normalization and turn map belong to this physical interface; clipping at a wall can reduce speed below 35 m/s. The absolute-position action has three continuous coordinates in $\mathcal{W}$. The outer loop supplies a norm-limited acceleration; the six raw self values provide the position and velocity needed for this reference interface. A position intent does not override collision or minimum free-flight speed. Only this model admits the planar compatibility setting on its physics page. All equations and dimensions here describe the native 3D instance.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M - 1) \times 11$ entity table ($V_{\max} = 120$) and unnormalized $[p_i, v_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\dim o_i = 11(M - 1) + 6, \quad \dim g = 11M. \quad (46)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = 1[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Damage reward and completion instance

Apply the Damage rule on page 3: finite asset HP stays at reset health; raw simultaneous damage is counted before clipping or overkill normalization.

$$\begin{aligned} h'_k &= h_k^0, & D'_k &= D_k + L_k, \\ R_t &= -\sum_k L_k, & d_t^{\text{term}} &= 1[\forall b \in \mathcal{B}_A : h'_b \leq 0]. \end{aligned} \quad (47)$$

Blue receives $-R_t$. Every team slot receives the full shared reward, including dead Red slots and later damage after Red extinction; summing duplicate teammate rewards changes the objective. Dead slots keep zero observations and ignored actions until global completion/truncation. At $\gamma = 1$, the Red return is $-\sum_k D_k$, with no Survival terminal bonus or optional Survival weights.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i | \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 11. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (48)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XIII. THEME 5 / FIXED WING / ACCELERATION / SURVIVAL

Rigid-body aerodynamic transition and navigation/actuator semantics.

A. Physical state for this interface

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state, health, role, side and replay records follow page 2.

B. Shared rigid-body transition

Use Newton–Euler (2) and RK4 (4) on page 2, with the wrench and controller specified here. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

C. Aerosonde aerodynamic wrench

Let $C_f = \text{diag}(1, -1, -1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin \text{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\hat{p} = bp_b/(2\bar{V})$, $\hat{q} = cq_b/(2\bar{V})$, $\hat{r} = br_b/(2\bar{V})$.

$$\begin{aligned} E_- &= e^{\text{clip}(-50(\alpha-.47), -500, 500)}, \\ E_+ &= e^{\text{clip}(50(\alpha+.47), -500, 500)}, \\ \sigma &= (1 + E_- + E_+)/[(1 + E_-)(1 + E_+)], \\ C_L &= (1 - \sigma)(.23 + 5.61\alpha) \\ &\quad + 2\sigma \text{sign}(\alpha) \sin^2 \alpha \cos \alpha + 7.95\hat{q} + .13\delta_e, \\ C_D &= .043 + \frac{(.23 + 5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e, \end{aligned} \quad (49)$$

$$\begin{aligned} C_Y &= -.98\beta + .075\delta_a + .19\delta_r, \\ C_\ell &= -.13\beta - .51\hat{p} + .25\hat{r} + .17\delta_a + .0024\delta_r, \\ C_m &= .0135 - 2.74\alpha - 38.21\hat{q} - .99\delta_e, \\ C_n &= .073\beta + .069\hat{p} - .095\hat{r} - .011\delta_a - .069\delta_r. \end{aligned}$$

$$F_B = C_f \left\{ \bar{Q} S \begin{bmatrix} -C_D \cos \alpha + C_L \sin \alpha \\ C_Y \\ -C_D \sin \alpha - C_L \cos \alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \quad (50)$$

$$M_B = C_f \left\{ \bar{Q} S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \quad (51)$$

D. Motor, propeller and equilibrium

The physical constants are $m = 11$ kg, $S = .55$ m², $b = 2.8956$ m, $c = .18994$ m, $\rho = 1.2682$ kg/m³,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T. \quad (52)$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145 \cdot 2\pi)$, the steady motor root is

$$\begin{aligned} A &= \rho D_p^5 C_{Q0}/(4\pi^2), \\ B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m, \\ C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0, \\ \Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}. \end{aligned} \quad (53)$$

Set $z = (D_p^4 \Omega_p^2/(4\pi^2), D_p^3 V \Omega_p/(2\pi), D_p^2 V^2)$,

$$\begin{aligned} T_p &= \rho(.09357, -.06044, -.1079)z, \\ Q_p &= \rho D_p(.005230, .004970, -.01664)z. \end{aligned} \quad (54)$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

E. Action and reference conversion

Choose $a_i \in [-1, 1]^3$ or one of the 27 unit primitives. Scale once to $\alpha_i = 5a_i$, then hold $v_d = v_i(t) + \Delta\alpha_i$. At each RK4 stage,

$$\begin{aligned} \chi_d &= \text{atan2}(v_{dy}, v_{dx}), \\ \gamma_d &= \text{atan2}(v_{dz}, \max(\|v_{d,xy}\|, 10^{-6})), \\ V_d &= \|v_d\|. \end{aligned} \quad (55)$$

For tiny horizontal reference speed, χ_d is current heading. Component-limited continuous intent need not have norm ≤ 5 .

F. Implemented reference controller

Define bank $\phi = \text{atan2}(R_{32}, R_{33})$, nose-up pitch $\theta = \arcsin R_{31}$, heading $\psi = \text{atan2}(R_{21}, R_{11})$. Actual course is $\chi = \text{atan2}(v_y, v_x)$, or ψ when horizontal speed is tiny; path $\gamma_f = \text{atan2}(v_z, \|v_{xy}\|)$ and $V_a = \|v\|$.

$$\begin{aligned} V_d &= \text{clip}(V_d, 18, 30), \\ \gamma_d &= \text{clip}(\gamma_d, -15^\circ, 15^\circ), \\ e_\chi &= \text{wrap}(\chi_d - \chi), \\ \bar{\phi} &= -\text{atan2}(\max(V_a, 18)e_\chi, g), \\ \phi_d &= \text{clip}(\bar{\phi}, -45^\circ, 45^\circ), \\ \bar{\theta} &= \theta^* + \gamma_d + .5(\gamma_d - \gamma_f), \\ \theta_d &= \text{clip}(\bar{\theta}, -20^\circ, 20^\circ), \\ \delta_e &= \delta_e^* - 2(\theta_d - \theta) + .25q_b, \\ \delta_a &= \delta_a^* + .5(\phi_d - \phi) - .08p_b, \\ r_d &= g \tan \phi_d / \max(V_a, 18), \\ \delta_r &= \delta_r^* + .8(\beta - \psi^*) + .1(r_b - r_d), \\ \tau &= \tau^* + .08(V_d - V_a) + .7 \sin \gamma_d. \end{aligned} \quad (56)$$

All angles use radians. The scheduled trim heading ψ^* is the lateral reference. Divide surfaces by 25° , map throttle to $2\tau - 1$ and clip the four outputs to $[-1, 1]$. These project gains/bounds do not clamp the physical flight envelope.

G. Integration, geometry and initialization

Use the common stage-feedback RK4 and quaternion normalization on page 2. Default $\lambda = .4$, $d_c = 8$ m, $(r_0, r_1) = (120, 240)$ m, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000]$ m. Clip path positions and the endpoint to scene bounds; remove endpoint outward velocity. The reset law ρ_0 uses the identical asset distribution, Red annulus, Blue rectangle, altitude and opening-separation law fully specified on page 17, with solved 30 m/s level trim. Only 3D is supported.

XIV. THEME 5 / FIXED WING / ACCELERATION / SURVIVAL

Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 13, combined with the shared event rules on page 2, defines F_C . The configuration selects **Fixed wing / Acceleration** and **Survival**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Fixed-wing event and scale instance

Use the constrained thirteen-value rigid state, $\lambda = .4$, $d_c = 8$ m and $(r_0, r_1) = (120, 240)$ m. Actor normalization uses $V_{\max} = 30$ m/s. Collision prediction uses synchronized, clipped RK4 path segments. Aerodynamic force, motor/propeller reaction torque, trim and attitude all affect navigation; they are preserved on the physics page. The acceleration action has three continuous coordinates or 27 unit primitives, scaled by 5 m/s². It creates the once-per-step held velocity reference; heading, path-angle and speed references then feed stage feedback. Reference bounds constrain the controller, rather than the actual flight envelope. Other agents' attitude and body rate are omitted from actor rows, even though those states determine their future motion. Self attitude/rate are retained in the raw own-state block.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 30$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\begin{aligned} \dim o_i &= 11(M-1) + 13, \\ \dim g &= 11M + 13N + 3. \end{aligned} \quad (57)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely

from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Survival reward and completion instance

Apply the Survival rule on page 3: asset HP decreases by raw L_k , and target loss has priority over Blue Attack elimination.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\text{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \quad (58)$$

Blue receives $-R_t$. Red elimination alone does not end the world. A nonpositive-health actor retires after its lethal final transition, while the team return and successor value continue until global natural completion. Explicit optional reward weights replace this native scalar by the formulation on page 4; they do not add a second terminal bonus.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i | \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 13. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (59)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XV. THEME 6 / FIXED WING / ACCELERATION / DAMAGE

Rigid-body aerodynamic transition and navigation/actuator semantics.

A. Physical state for this interface

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state, health, role, side and replay records follow page 2.

B. Shared rigid-body transition

Use Newton–Euler (2) and RK4 (4) on page 2, with the wrench and controller specified here. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

C. Aerosonde aerodynamic wrench

Let $C_f = \text{diag}(1, -1, -1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin \text{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\hat{p} = b p_b/(2\bar{V})$, $\hat{q} = c q_b/(2\bar{V})$, $\hat{r} = b r_b/(2\bar{V})$.

$$\begin{aligned} E_- &= e^{\text{clip}(-50(\alpha-.47), -500, 500)}, \\ E_+ &= e^{\text{clip}(50(\alpha+.47), -500, 500)}, \\ \sigma &= (1 + E_- + E_+)/[(1 + E_-)(1 + E_+)], \\ C_L &= (1 - \sigma)(.23 + 5.61\alpha) \\ &\quad + 2\sigma \text{sign}(\alpha) \sin^2 \alpha \cos \alpha + 7.95\hat{q} + .13\delta_e, \\ C_D &= .043 + \frac{(.23 + 5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e, \\ C_Y &= -.98\beta + .075\delta_a + .19\delta_r, \\ C_\ell &= -.13\beta - .51\hat{p} + .25\hat{r} + .17\delta_a + .0024\delta_r, \\ C_m &= .0135 - 2.74\alpha - 38.21\hat{q} - .99\delta_e, \\ C_n &= .073\beta + .069\hat{p} - .095\hat{r} - .011\delta_a - .069\delta_r. \end{aligned} \quad (60)$$

$$F_B = C_f \left\{ \bar{Q} S \begin{bmatrix} -C_D \cos \alpha + C_L \sin \alpha \\ C_Y \\ -C_D \sin \alpha - C_L \cos \alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \quad (61)$$

$$M_B = C_f \left\{ \bar{Q} S \begin{bmatrix} b C_\ell \\ c C_m \\ b C_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \quad (62)$$

D. Motor, propeller and equilibrium

The physical constants are $m = 11$ kg, $S = .55$ m², $b = 2.8956$ m, $c = .18994$ m, $\rho = 1.2682$ kg/m³,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T. \quad (63)$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145 \cdot 2\pi)$, the steady motor root is

$$\begin{aligned} A &= \rho D_p^5 C_{Q0}/(4\pi^2), \\ B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m, \\ C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0, \\ \Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}. \end{aligned} \quad (64)$$

Set $z = (D_p^4 \Omega_p^2/(4\pi^2), D_p^3 V \Omega_p/(2\pi), D_p^2 V^2)$,

$$\begin{aligned} T_p &= \rho(.09357, -.06044, -.1079)z, \\ Q_p &= \rho D_p(.005230, .004970, -.01664)z. \end{aligned} \quad (65)$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

E. Action and reference conversion

Choose $a_i \in [-1, 1]^3$ or one of the 27 unit primitives. Scale once to $\alpha_i = 5a_i$, then hold $v_d = v_i(t) + \Delta\alpha_i$. At each RK4 stage,

$$\begin{aligned} \chi_d &= \text{atan2}(v_{dy}, v_{dx}), \\ \gamma_d &= \text{atan2}(v_{dz}, \max(\|v_{d,xy}\|, 10^{-6})), \\ V_d &= \|v_d\|. \end{aligned} \quad (66)$$

For tiny horizontal reference speed, χ_d is current heading. Component-limited continuous intent need not have norm ≤ 5 .

F. Implemented reference controller

Define bank $\phi = \text{atan2}(R_{32}, R_{33})$, nose-up pitch $\theta = \arcsin R_{31}$, heading $\psi = \text{atan2}(R_{21}, R_{11})$. Actual course is $\chi = \text{atan2}(v_y, v_x)$, or ψ when horizontal speed is tiny; path $\gamma_f = \text{atan2}(v_z, \|v_{xy}\|)$ and $V_a = \|v\|$.

$$\begin{aligned} V_d &= \text{clip}(V_d, 18, 30), \\ \gamma_d &= \text{clip}(\gamma_d, -15^\circ, 15^\circ), \\ e_\chi &= \text{wrap}(\chi_d - \chi), \\ \bar{\phi} &= -\text{atan2}(\max(V_a, 18)e_\chi, g), \\ \phi_d &= \text{clip}(\bar{\phi}, -45^\circ, 45^\circ), \\ \bar{\theta} &= \theta^* + \gamma_d + .5(\gamma_d - \gamma_f), \\ \theta_d &= \text{clip}(\bar{\theta}, -20^\circ, 20^\circ), \\ \delta_e &= \delta_e^* - 2(\theta_d - \theta) + .25q_b, \\ \delta_a &= \delta_a^* + .5(\phi_d - \phi) - .08p_b, \\ r_d &= g \tan \phi_d / \max(V_a, 18), \\ \delta_r &= \delta_r^* + .8(\beta - \psi^*) + .1(r_b - r_d), \\ \tau &= \tau^* + .08(V_d - V_a) + .7 \sin \gamma_d. \end{aligned} \quad (67)$$

All angles use radians. The scheduled trim heading ψ^* is the lateral reference. Divide surfaces by 25° , map throttle to $2\tau - 1$ and clip the four outputs to $[-1, 1]$. These project gains/bounds do not clamp the physical flight envelope.

G. Integration, geometry and initialization

Use the common stage-feedback RK4 and quaternion normalization on page 2. Default $\lambda = .4$, $d_c = 8$ m, $(r_0, r_1) = (120, 240)$ m, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000]$ m. Clip path positions and the endpoint to scene bounds; remove endpoint outward velocity. The reset law ρ_0 uses the identical asset distribution, Red annulus, Blue rectangle, altitude and opening-separation law fully specified on page 17, with solved 30 m/s level trim. Only 3D is supported.

XVI. THEME 6 / FIXED WING / ACCELERATION / DAMAGE
Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 15, combined with the shared event rules on page 2, defines F_C . The configuration selects **Fixed wing / Acceleration and Damage**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Fixed-wing event and scale instance

Use the constrained thirteen-value rigid state, $\lambda = .4$, $d_c = 8$ m and $(r_0, r_1) = (120, 240)$ m. Actor normalization uses $V_{\max} = 30$ m/s. Collision prediction uses synchronized, clipped RK4 path segments. Aerodynamic force, motor/propeller reaction torque, trim and attitude all affect navigation; they are preserved on the physics page. The acceleration action has three continuous coordinates or 27 unit primitives, scaled by 5 m/s². It creates the once-per-step held velocity reference; heading, path-angle and speed references then feed stage feedback. Reference bounds constrain the controller, rather than the actual flight envelope. Other agents' attitude and body rate are omitted from actor rows, even though those states determine their future motion. Self attitude/rate are retained in the raw own-state block.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 30$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\begin{aligned} \dim o_i &= 11(M-1) + 13, \\ \dim g &= 11M + 13N + 3. \end{aligned} \quad (68)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = 1[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely

from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Damage reward and completion instance

Apply the Damage rule on page 3: finite asset HP stays at reset health; raw simultaneous damage is counted before clipping or overkill normalization.

$$\begin{aligned} h'_k &= h_k^0, & D'_k &= D_k + L_k, \\ R_t &= -\sum_k L_k, & d_t^{\text{term}} &= 1[\forall b \in \mathcal{B}_A : h'_b \leq 0]. \end{aligned} \quad (69)$$

Blue receives $-R_t$. Every team slot receives the full shared reward, including dead Red slots and later damage after Red extinction; summing duplicate teammate rewards changes the objective. Dead slots keep zero observations and ignored actions until global completion/truncation. At $\gamma = 1$, the Red return is $-\sum_k D_k$, with no Survival terminal bonus or optional Survival weights.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i | \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 15. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (70)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XVII. THEME 7 / FIXED WING / ACTUATOR / SURVIVAL

Rigid-body aerodynamic transition and navigation/actuator semantics.

A. Physical state for this interface

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state, health, role, side and replay records follow page 2.

B. Shared rigid-body transition

Use Newton–Euler (2) and RK4 (4) on page 2, with the wrench and controller specified here. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

C. Aerosonde aerodynamic wrench

Let $C_f = \text{diag}(1, -1, -1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin \text{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\hat{p} = bp_b/(2\bar{V})$, $\hat{q} = cq_b/(2\bar{V})$, $\hat{r} = br_b/(2\bar{V})$.

$$\begin{aligned} E_- &= e^{\text{clip}(-50(\alpha-.47), -500, 500)}, \\ E_+ &= e^{\text{clip}(50(\alpha+.47), -500, 500)}, \\ \sigma &= (1 + E_- + E_+)/[(1 + E_-)(1 + E_+)], \\ C_L &= (1 - \sigma)(.23 + 5.61\alpha) \\ &\quad + 2\sigma \text{sign}(\alpha) \sin^2 \alpha \cos \alpha + 7.95\hat{q} + .13\delta_e, \\ C_D &= .043 + \frac{(.23 + 5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e, \end{aligned} \quad (71)$$

$$\begin{aligned} C_Y &= -.98\beta + .075\delta_a + .19\delta_r, \\ C_\ell &= -.13\beta - .51\hat{p} + .25\hat{r} + .17\delta_a + .0024\delta_r, \\ C_m &= .0135 - 2.74\alpha - 38.21\hat{q} - .99\delta_e, \\ C_n &= .073\beta + .069\hat{p} - .095\hat{r} - .011\delta_a - .069\delta_r. \end{aligned}$$

$$F_B = C_f \left\{ \bar{Q} S \begin{bmatrix} -C_D \cos \alpha + C_L \sin \alpha \\ C_Y \\ -C_D \sin \alpha - C_L \cos \alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \quad (72)$$

$$M_B = C_f \left\{ \bar{Q} S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \quad (73)$$

D. Motor, propeller and equilibrium

The physical constants are $m = 11$ kg, $S = .55$ m², $b = 2.8956$ m, $c = .18994$ m, $\rho = 1.2682$ kg/m³,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T. \quad (74)$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145 \cdot 2\pi)$, the steady motor root is

$$\begin{aligned} A &= \rho D_p^5 C_{Q0}/(4\pi^2), \\ B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m, \\ C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0, \\ \Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}. \end{aligned} \quad (75)$$

Set $z = (D_p^4 \Omega_p^2/(4\pi^2), D_p^3 V \Omega_p/(2\pi), D_p^2 V^2)$,

$$\begin{aligned} T_p &= \rho(.09357, -.06044, -.1079)z, \\ Q_p &= \rho D_p(.005230, .004970, -.01664)z. \end{aligned} \quad (76)$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

E. Direct actuator action

Every living agent selects $a_i \in [-1, 1]^4$, ordered elevator, aileron, rudder, throttle:

$$(\delta_e, \delta_a, \delta_r) = 25^\circ(a_1, a_2, a_3), \quad \tau = (a_4 + 1)/2. \quad (77)$$

The actuator vector is constant for the external step and feeds the aerodynamic/motor equations directly. There is no automatic stabilization or navigation controller. In particular $a = 0$ is neutral surfaces with half throttle, not solved trim. Automatic fire still uses the same separate task rules.

F. Numerical and boundary rules

Use the synchronized 100-substep RK4 map (4) and clipped-path boundary rule on page 2. Hold the external command for 1 s; stage feedback is recomputed only for intent interfaces. Direct actuators remain constant.

G. Geometry and initialization

Default $\lambda = .4$, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000]$ m, $d_c = 8$ m, $(r_0, r_1) = (120, 240)$ m. Targets use the scaled region $[-920, -760] \times [-480, 480] \times [200, 600]$ m. Red uses an area-uniform XY annulus $[320, 880]$ m around a sampled target; Blue uses $x \in [0, 1000]$ and full scene y ; agent altitude lies in $[80, 600]$ m. UAV openings reject distances $\leq \max(30\lambda, d_c + 10^{-6})$. Both teams reset at solved 30 m/s level flight; Blue heads west, Red east under random initialization. Only 3D is supported.

XVIII. THEME 7 / FIXED WING / ACTUATOR / SURVIVAL

Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 17, combined with the shared event rules on page 2, defines F_C . The configuration selects **Fixed wing / Actuator** and **Survival**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Fixed-wing event and scale instance

Use the constrained thirteen-value rigid state, $\lambda = .4$, $d_c = 8$ m and $(r_0, r_1) = (120, 240)$ m. Actor normalization uses $V_{\max} = 30$ m/s. Collision prediction uses synchronized, clipped RK4 path segments. Aerodynamic force, motor/propeller reaction torque, trim and attitude all affect navigation; they are preserved on the physics page. The actuator action has four continuous coordinates in $[-1, 1]$, ordered elevator, aileron, rudder, throttle. It acts directly on the wrench for the whole step. Neither the trim schedule nor the navigation controller repairs destabilizing actions. Other agents' attitude and body rate are omitted from actor rows, even though those states determine their future motion. Self attitude/rate are retained in the raw own-state block.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 30$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Actuator controls do not add an observation block. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\begin{aligned} \dim o_i &= 11(M-1) + 13, \\ \dim g &= 11M + 13N + 3. \end{aligned} \quad (78)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = 1[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Survival reward and completion instance

Apply the Survival rule on page 3: asset HP decreases by raw L_k , and target loss has priority over Blue Attack elimination.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\text{term}} &= 1[z(s') \neq 0]. \end{aligned} \quad (79)$$

Blue receives $-R_t$. Red elimination alone does not end the world. A nonpositive-health actor retires after its lethal final transition, while the team return and successor value continue until global natural completion. Explicit optional reward weights replace this native scalar by the formulation on page 4; they do not add a second terminal bonus.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i | \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 17. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (80)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XIX. THEME 8 / FIXED WING / ACTUATOR / DAMAGE

Rigid-body aerodynamic transition and navigation/actuator semantics.

A. Physical state for this interface

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state, health, role, side and replay records follow page 2.

B. Shared rigid-body transition

Use Newton–Euler (2) and RK4 (4) on page 2, with the wrench and controller specified here. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

C. Aerosonde aerodynamic wrench

Let $C_f = \text{diag}(1, -1, -1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin \text{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\hat{p} = b p_b/(2\bar{V})$, $\hat{q} = c q_b/(2\bar{V})$, $\hat{r} = b r_b/(2\bar{V})$.

$$\begin{aligned} E_- &= e^{\text{clip}(-50(\alpha-.47), -500, 500)}, \\ E_+ &= e^{\text{clip}(50(\alpha+.47), -500, 500)}, \\ \sigma &= (1 + E_- + E_+)/[(1 + E_-)(1 + E_+)], \\ C_L &= (1 - \sigma)(.23 + 5.61\alpha) \\ &\quad + 2\sigma \text{sign}(\alpha) \sin^2 \alpha \cos \alpha + 7.95\hat{q} + .13\delta_e, \end{aligned} \quad (81)$$

$$C_D = .043 + \frac{(.23 + 5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e,$$

$$C_Y = -.98\beta + .075\delta_a + .19\delta_r,$$

$$C_\ell = -.13\beta - .51\hat{p} + .25\hat{r} + .17\delta_a + .0024\delta_r,$$

$$C_m = .0135 - 2.74\alpha - 38.21\hat{q} - .99\delta_e,$$

$$C_n = .073\beta + .069\hat{p} - .095\hat{r} - .011\delta_a - .069\delta_r.$$

$$F_B = C_f \left\{ \bar{Q} S \begin{bmatrix} -C_D \cos \alpha + C_L \sin \alpha \\ C_Y \\ -C_D \sin \alpha - C_L \cos \alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \quad (82)$$

$$M_B = C_f \left\{ \bar{Q} S \begin{bmatrix} b C_\ell \\ c C_m \\ b C_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \quad (83)$$

D. Motor, propeller and equilibrium

The physical constants are $m = 11$ kg, $S = .55$ m², $b = 2.8956$ m, $c = .18994$ m, $\rho = 1.2682$ kg/m³,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T. \quad (84)$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145 \cdot 2\pi)$, the steady motor root is

$$\begin{aligned} A &= \rho D_p^5 C_{Q0}/(4\pi^2), \\ B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m, \\ C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0, \\ \Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}. \end{aligned} \quad (85)$$

$$\text{Set } z = (D_p^4 \Omega_p^2/(4\pi^2), D_p^3 V \Omega_p/(2\pi), D_p^2 V^2),$$

$$\begin{aligned} T_p &= \rho(.09357, -.06044, -.1079)z, \\ Q_p &= \rho D_p(.005230, .004970, -.01664)z. \end{aligned} \quad (86)$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

E. Direct actuator action

Every living agent selects $a_i \in [-1, 1]^4$, ordered elevator, aileron, rudder, throttle:

$$(\delta_e, \delta_a, \delta_r) = 25^\circ(a_1, a_2, a_3), \quad \tau = (a_4 + 1)/2. \quad (87)$$

The actuator vector is constant for the external step and feeds the aerodynamic/motor equations directly. There is no automatic stabilization or navigation controller. In particular $a = 0$ is neutral surfaces with half throttle, not solved trim. Automatic fire still uses the same separate task rules.

F. Numerical and boundary rules

Use the synchronized 100-substep RK4 map (4) and clipped-path boundary rule on page 2. Hold the external command for 1 s; stage feedback is recomputed only for intent interfaces. Direct actuators remain constant.

G. Geometry and initialization

Default $\lambda = .4$, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000]$ m, $d_c = 8$ m, $(r_0, r_1) = (120, 240)$ m. Targets use the scaled region $[-920, -760] \times [-480, 480] \times [200, 600]$ m. Red uses an area-uniform XY annulus $[320, 880]$ m around a sampled target; Blue uses $x \in [0, 1000]$ and full scene y ; agent altitude lies in $[80, 600]$ m. UAV openings reject distances $\leq \max(30\lambda, d_c + 10^{-6})$. Both teams reset at solved 30 m/s level flight; Blue heads west, Red east under random initialization. Only 3D is supported.

XX. THEME 8 / FIXED WING / ACTUATOR / DAMAGE

Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 19, combined with the shared event rules on page 2, defines F_C . The configuration selects **Fixed wing / Actuator** and **Damage**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Fixed-wing event and scale instance

Use the constrained thirteen-value rigid state, $\lambda = .4$, $d_c = 8$ m and $(r_0, r_1) = (120, 240)$ m. Actor normalization uses $V_{\max} = 30$ m/s. Collision prediction uses synchronized, clipped RK4 path segments. Aerodynamic force, motor/propeller reaction torque, trim and attitude all affect navigation; they are preserved on the physics page. The actuator action has four continuous coordinates in $[-1, 1]$, ordered elevator, aileron, rudder, throttle. It acts directly on the wrench for the whole step. Neither the trim schedule nor the navigation controller repairs destabilizing actions. Other agents' attitude and body rate are omitted from actor rows, even though those states determine their future motion. Self attitude/rate are retained in the raw own-state block.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 30$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\begin{aligned} \dim o_i &= 11(M-1) + 13, \\ \dim g &= 11M + 13N + 3. \end{aligned} \quad (88)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = 1[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Damage reward and completion instance

Apply the Damage rule on page 3: finite asset HP stays at reset health; raw simultaneous damage is counted before clipping or overkill normalization.

$$\begin{aligned} h'_k &= h_k^0, & D'_k &= D_k + L_k, \\ R_t &= -\sum_k L_k, & d_t^{\text{term}} &= 1[\forall b \in \mathcal{B}_A : h'_b \leq 0]. \end{aligned} \quad (89)$$

Blue receives $-R_t$. Every team slot receives the full shared reward, including dead Red slots and later damage after Red extinction; summing duplicate teammate rewards changes the objective. Dead slots keep zero observations and ignored actions until global completion/truncation. At $\gamma = 1$, the Red return is $-\sum_k D_k$, with no Survival terminal bonus or optional Survival weights.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i | \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 19. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (90)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XXI. THEME 9 / FIXED WING / POSITION / SURVIVAL

Rigid-body aerodynamic transition and navigation/actuator semantics.

A. Physical state for this interface

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state, health, role, side and replay records follow page 2.

B. Shared rigid-body transition

Use Newton–Euler (2) and RK4 (4) on page 2, with the wrench and controller specified here. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

C. Aerosonde aerodynamic wrench

Let $C_f = \text{diag}(1, -1, -1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin \text{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\hat{p} = bp_b/(2\bar{V})$, $\hat{q} = cq_b/(2\bar{V})$, $\hat{r} = br_b/(2\bar{V})$.

$$\begin{aligned} E_- &= e^{\text{clip}(-50(\alpha-.47), -500, 500)}, \\ E_+ &= e^{\text{clip}(50(\alpha+.47), -500, 500)}, \\ \sigma &= (1 + E_- + E_+)/[(1 + E_-)(1 + E_+)], \\ C_L &= (1 - \sigma)(.23 + 5.61\alpha) \\ &\quad + 2\sigma \text{sign}(\alpha) \sin^2 \alpha \cos \alpha + 7.95\hat{q} + .13\delta_e, \end{aligned} \quad (91)$$

$$\begin{aligned} C_D &= .043 + \frac{(.23 + 5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e, \\ C_Y &= -.98\beta + .075\delta_a + .19\delta_r, \\ C_\ell &= -.13\beta - .51\hat{p} + .25\hat{r} + .17\delta_a + .0024\delta_r, \\ C_m &= .0135 - 2.74\alpha - 38.21\hat{q} - .99\delta_e, \\ C_n &= .073\beta + .069\hat{p} - .095\hat{r} - .011\delta_a - .069\delta_r. \end{aligned}$$

$$F_B = C_f \left\{ \bar{Q} S \begin{bmatrix} -C_D \cos \alpha + C_L \sin \alpha \\ C_Y \\ -C_D \sin \alpha - C_L \cos \alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \quad (92)$$

$$M_B = C_f \left\{ \bar{Q} S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \quad (93)$$

D. Motor, propeller and equilibrium

The physical constants are $m = 11$ kg, $S = .55$ m², $b = 2.8956$ m, $c = .18994$ m, $\rho = 1.2682$ kg/m³,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T. \quad (94)$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145 \cdot 2\pi)$, the steady motor root is

$$\begin{aligned} A &= \rho D_p^5 C_{Q0}/(4\pi^2), \\ B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m, \\ C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0, \\ \Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}. \end{aligned} \quad (95)$$

Set $z = (D_p^4 \Omega_p^2/(4\pi^2), D_p^3 V \Omega_p/(2\pi), D_p^2 V^2)$,

$$\begin{aligned} T_p &= \rho(.09357, -.06044, -.1079)z, \\ Q_p &= \rho D_p(.005230, .004970, -.01664)z. \end{aligned} \quad (96)$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

E. Absolute position action

Choose $a_i = p_d \in \mathcal{W}$ in metres. At every stage, $d = p_d - p$ determines

$$\begin{aligned} \chi_d &= \text{atan2}(d_y, d_x), \\ \gamma_d &= \text{atan2}(d_z, \max(\|d_{xy}\|, 30)), \quad V_d = 30. \end{aligned} \quad (97)$$

Tiny horizontal displacement uses current heading. The aircraft flies toward and through the target; it cannot hover there.

F. Implemented reference controller

Define bank $\phi = \text{atan2}(R_{32}, R_{33})$, nose-up pitch $\theta = \arcsin R_{31}$, heading $\psi = \text{atan2}(R_{21}, R_{11})$. Actual course is $\chi = \text{atan2}(v_y, v_x)$, or ψ when horizontal speed is tiny; path $\gamma_f = \text{atan2}(v_z, \|v_{xy}\|)$ and $V_a = \|v\|$.

$$\begin{aligned} V_d &= \text{clip}(V_d, 18, 30), \\ \gamma_d &= \text{clip}(\gamma_d, -15^\circ, 15^\circ), \\ e_\chi &= \text{wrap}(\chi_d - \chi), \\ \bar{\phi} &= -\text{atan2}(\max(V_a, 18)e_\chi, g), \\ \phi_d &= \text{clip}(\bar{\phi}, -45^\circ, 45^\circ), \\ \bar{\theta} &= \theta^* + \gamma_d + .5(\gamma_d - \gamma_f), \\ \theta_d &= \text{clip}(\bar{\theta}, -20^\circ, 20^\circ), \\ \delta_e &= \delta_e^* - 2(\theta_d - \theta) + .25q_b, \\ \delta_a &= \delta_a^* + .5(\phi_d - \phi) - .08p_b, \\ r_d &= g \tan \phi_d / \max(V_a, 18), \\ \delta_r &= \delta_r^* + .8(\beta - \psi^*) + .1(r_b - r_d), \\ \tau &= \tau^* + .08(V_d - V_a) + .7 \sin \gamma_d. \end{aligned} \quad (98)$$

All angles use radians. The scheduled trim heading ψ^* is the lateral reference. Divide surfaces by 25° , map throttle to $2\tau - 1$ and clip the four outputs to $[-1, 1]$. These project gains/bounds do not clamp the physical flight envelope.

G. Integration, geometry and initialization

Use the common stage-feedback RK4 and quaternion normalization on page 2. Default $\lambda = .4$, $d_c = 8$ m, $(r_0, r_1) = (120, 240)$ m, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000]$ m. Clip path positions and the endpoint to scene bounds; remove endpoint outward velocity. The reset law ρ_0 uses the identical asset distribution, Red annulus, Blue rectangle, altitude and opening-separation law fully specified on page 17, with solved 30 m/s level trim. Only 3D is supported.

XXII. THEME 9 / FIXED WING / POSITION / SURVIVAL

Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 21, combined with the shared event rules on page 2, defines F_C . The configuration selects **Fixed wing / Position** and **Survival**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Fixed-wing event and scale instance

Use the constrained thirteen-value rigid state, $\lambda = .4$, $d_c = 8$ m and $(r_0, r_1) = (120, 240)$ m. Actor normalization uses $V_{\max} = 30$ m/s. Collision prediction uses synchronized, clipped RK4 path segments. Aerodynamic force, motor/propeller reaction torque, trim and attitude all affect navigation; they are preserved on the physics page. The position action has three continuous coordinates in $\mathcal{W}$. Stage position determines heading and climb reference with 30 m/s speed demand. The same scheduled-trim controller flies toward and through the point; the vehicle cannot hover. Other agents' attitude and body rate are omitted from actor rows, even though those states determine their future motion. Self attitude/rate are retained in the raw own-state block.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 30$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\begin{aligned} \dim o_i &= 11(M-1) + 13, \\ \dim g &= 11M + 13N + 3. \end{aligned} \quad (99)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Survival reward and completion instance

Apply the Survival rule on page 3: asset HP decreases by raw L_k , and target loss has priority over Blue Attack elimination.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\text{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \quad (100)$$

Blue receives $-R_t$. Red elimination alone does not end the world. A nonpositive-health actor retires after its lethal final transition, while the team return and successor value continue until global natural completion. Explicit optional reward weights replace this native scalar by the formulation on page 4; they do not add a second terminal bonus.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i \mid \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 21. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (101)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XXIII. THEME 10 / FIXED WING / POSITION / DAMAGE
Rigid-body aerodynamic transition and navigation/actuator semantics.

A. Physical state for this interface

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state, health, role, side and replay records follow page 2.

B. Shared rigid-body transition

Use Newton–Euler (2) and RK4 (4) on page 2, with the wrench and controller specified here. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

C. Aerosonde aerodynamic wrench

Let $C_f = \text{diag}(1, -1, -1)$ and $(u, v_b, w) = C_f R^T v$, $(p_b, q_b, r_b) = C_f \omega$. Define $V = \|(u, v_b, w)\|$, $\bar{V} = \max(V, 10^{-6})$, $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin \text{clip}(v_b/\bar{V}, -1, 1)$, $\bar{Q} = \rho V^2/2$, $\hat{p} = bp_b/(2\bar{V})$, $\hat{q} = cq_b/(2\bar{V})$, $\hat{r} = br_b/(2\bar{V})$.

$$\begin{aligned} E_- &= e^{\text{clip}(-50(\alpha-.47), -500, 500)}, \\ E_+ &= e^{\text{clip}(50(\alpha+.47), -500, 500)}, \\ \sigma &= (1 + E_- + E_+)/[(1 + E_-)(1 + E_+)], \\ C_L &= (1 - \sigma)(.23 + 5.61\alpha) \\ &\quad + 2\sigma \text{sign}(\alpha) \sin^2 \alpha \cos \alpha + 7.95\hat{q} + .13\delta_e, \end{aligned} \quad (102)$$

$$\begin{aligned} C_D &= .043 + \frac{(.23 + 5.61\alpha)^2}{\pi(.9)(b^2/S)} + .0135\delta_e, \\ C_Y &= -.98\beta + .075\delta_a + .19\delta_r, \\ C_\ell &= -.13\beta - .51\hat{p} + .25\hat{r} + .17\delta_a + .0024\delta_r, \\ C_m &= .0135 - 2.74\alpha - 38.21\hat{q} - .99\delta_e, \\ C_n &= .073\beta + .069\hat{p} - .095\hat{r} - .011\delta_a - .069\delta_r. \end{aligned}$$

$$F_B = C_f \left\{ \bar{Q} S \begin{bmatrix} -C_D \cos \alpha + C_L \sin \alpha \\ C_Y \\ -C_D \sin \alpha - C_L \cos \alpha \end{bmatrix} + \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \right\}, \quad (103)$$

$$M_B = C_f \left\{ \bar{Q} S \begin{bmatrix} bC_\ell \\ cC_m \\ bC_n \end{bmatrix} - \begin{bmatrix} Q_p \\ 0 \\ 0 \end{bmatrix} \right\}. \quad (104)$$

D. Motor, propeller and equilibrium

The physical constants are $m = 11$ kg, $S = .55$ m², $b = 2.8956$ m, $c = .18994$ m, $\rho = 1.2682$ kg/m³,

$$J = C_f \begin{bmatrix} .8244 & 0 & -.1204 \\ 0 & 1.135 & 0 \\ -.1204 & 0 & 1.759 \end{bmatrix} C_f^T. \quad (105)$$

With $D_p = .508$, $R_m = .042$, $I_0 = 1.5$, $V_m = 44.4$, $k_m = 60/(145 \cdot 2\pi)$, the steady motor root is

$$\begin{aligned} A &= \rho D_p^5 C_{Q0}/(4\pi^2), \\ B &= \rho D_p^4 C_{Q1} V/(2\pi) + k_m^2/R_m, \\ C &= \rho D_p^3 C_{Q2} V^2 - k_m V_m \tau/R_m + k_m I_0, \\ \Omega_p &= \max\{0, (-B + \sqrt{\max(0, B^2 - 4AC)})/(2A)\}. \end{aligned} \quad (106)$$

Set $z = (D_p^4 \Omega_p^2/(4\pi^2), D_p^3 V \Omega_p/(2\pi), D_p^2 V^2)$,

$$\begin{aligned} T_p &= \rho(.09357, -.06044, -.1079)z, \\ Q_p &= \rho D_p(.005230, .004970, -.01664)z. \end{aligned} \quad (107)$$

Reference trim $(x^*(V_d), u^*(V_d))$ solves all six equations $\dot{v} = 0$, $\dot{\omega} = 0$, including propeller reaction torque. Unknowns are pitch, heading, elevator, aileron, rudder and throttle, at velocity $(V_d, 0, 0)$. The reset uses the 30 m/s level trim rotated to the initial heading. Mass/inertia are not scaled with the scene [1].

E. Absolute position action

Choose $a_i = p_d \in \mathcal{W}$ in metres. At every stage, $d = p_d - p$ determines

$$\begin{aligned} \chi_d &= \text{atan2}(d_y, d_x), \\ \gamma_d &= \text{atan2}(d_z, \max(\|d_{xy}\|, 30)), \quad V_d = 30. \end{aligned} \quad (108)$$

Tiny horizontal displacement uses current heading. The aircraft flies toward and through the target; it cannot hover there.

F. Implemented reference controller

Define bank $\phi = \text{atan2}(R_{32}, R_{33})$, nose-up pitch $\theta = \arcsin R_{31}$, heading $\psi = \text{atan2}(R_{21}, R_{11})$. Actual course is $\chi = \text{atan2}(v_y, v_x)$, or ψ when horizontal speed is tiny; path $\gamma_f = \text{atan2}(v_z, \|v_{xy}\|)$ and $V_a = \|v\|$.

$$\begin{aligned} V_d &= \text{clip}(V_d, 18, 30), \\ \gamma_d &= \text{clip}(\gamma_d, -15^\circ, 15^\circ), \\ e_\chi &= \text{wrap}(\chi_d - \chi), \\ \bar{\phi} &= -\text{atan2}(\max(V_a, 18)e_\chi, g), \\ \phi_d &= \text{clip}(\bar{\phi}, -45^\circ, 45^\circ), \\ \bar{\theta} &= \theta^* + \gamma_d + .5(\gamma_d - \gamma_f), \\ \theta_d &= \text{clip}(\bar{\theta}, -20^\circ, 20^\circ), \\ \delta_e &= \delta_e^* - 2(\theta_d - \theta) + .25q_b, \\ \delta_a &= \delta_a^* + .5(\phi_d - \phi) - .08p_b, \\ r_d &= g \tan \phi_d / \max(V_a, 18), \\ \delta_r &= \delta_r^* + .8(\beta - \psi^*) + .1(r_b - r_d), \\ \tau &= \tau^* + .08(V_d - V_a) + .7 \sin \gamma_d. \end{aligned} \quad (109)$$

All angles use radians. The scheduled trim heading ψ^* is the lateral reference. Divide surfaces by 25° , map throttle to $2\tau - 1$ and clip the four outputs to $[-1, 1]$. These project gains/bounds do not clamp the physical flight envelope.

G. Integration, geometry and initialization

Use the common stage-feedback RK4 and quaternion normalization on page 2. Default $\lambda = .4$, $d_c = 8$ m, $(r_0, r_1) = (120, 240)$ m, $\mathcal{W} = [-1000, 1000]^2 \times [0, 1000]$ m. Clip path positions and the endpoint to scene bounds; remove endpoint outward velocity. The reset law ρ_0 uses the identical asset distribution, Red annulus, Blue rectangle, altitude and opening-separation law fully specified on page 17, with solved 30 m/s level trim. Only 3D is supported.

XXIV. THEME 10 / FIXED WING / POSITION / DAMAGE

Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 23, combined with the shared event rules on page 2, defines F_C . The configuration selects **Fixed wing / Position** and **Damage**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Fixed-wing event and scale instance

Use the constrained thirteen-value rigid state, $\lambda = .4$, $d_c = 8$ m and $(r_0, r_1) = (120, 240)$ m. Actor normalization uses $V_{\max} = 30$ m/s. Collision prediction uses synchronized, clipped RK4 path segments. Aerodynamic force, motor/propeller reaction torque, trim and attitude all affect navigation; they are preserved on the physics page. The position action has three continuous coordinates in $\mathcal{W}$. Stage position determines heading and climb reference with 30 m/s speed demand. The same scheduled-trim controller flies toward and through the point; the vehicle cannot hover. Other agents' attitude and body rate are omitted from actor rows, even though those states determine their future motion. Self attitude/rate are retained in the raw own-state block.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 30$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\begin{aligned} \dim o_i &= 11(M-1) + 13, \\ \dim g &= 11M + 13N + 3. \end{aligned} \quad (110)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Damage reward and completion instance

Apply the Damage rule on page 3: finite asset HP stays at reset health; raw simultaneous damage is counted before clipping or overkill normalization.

$$\begin{aligned} h'_k &= h_k^0, & D'_k &= D_k + L_k, \\ R_t &= -\sum_k L_k, & d_t^{\text{term}} &= \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0]. \end{aligned} \quad (111)$$

Blue receives $-R_t$. Every team slot receives the full shared reward, including dead Red slots and later damage after Red extinction; summing duplicate teammate rewards changes the objective. Dead slots keep zero observations and ignored actions until global completion/truncation. At $\gamma = 1$, the Red return is $-\sum_k D_k$, with no Survival terminal bonus or optional Survival weights.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i | \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 23. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (112)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XXV. THEME 11 / QUADROTOR / ACCELERATION / SURVIVAL

Rigid-body thrust dynamics, reference feedback and feasible actuation.

A. Physical state for this interface

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state, health, role, side and replay records follow page 2.

B. Shared rigid-body transition

Use Newton–Euler (2) and RK4 (4) on page 2, with the wrench and controller specified here. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

C. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34$ kg, $J = \text{diag}(.0820, .0845, .1377)$ kg m², arm $l = .315$ m and yaw ratio $c_\tau = .008004$ m [2],

$$f_{\max} = 2.5mg/4, \quad f_j = f_{\max}u_j, \quad (113)$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \quad (114)$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \quad (115)$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z) . Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

D. Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400]$ m, $d_c = 3.2$ m, $(r_0, r_1) = (48, 96)$ m and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240]$ m or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352]$ m around a uniformly chosen asset, clipped to scene bounds; Blue is uniform over $[0, 400] \times [-400, 400]$ m. UAV altitude is uniform in $[32, 240]$ m.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6})$ m of any previously placed entity; later collisions are allowed. Only 3D is supported.

E. Acceleration intent

Each living agent chooses either $a_i \in [-1, 1]^3$ or an index into

$$\mathcal{V} = \{0\} \cup \{z/\|z\| : z \in \{-1, 0, 1\}^3 \setminus \{0\}\}, \quad (116)$$

giving 27 discrete primitives, zero first. The physical intent is $\alpha_i = 6a_i$ m/s². Continuous inputs have component bounds, so diagonal intents can have norm $\sqrt{3}6$; discrete nonzero intents have norm 6. 6 is a control scale, not a clamp on actual rigid-body acceleration. The external intent establishes $v_d = v(t) + 6\Delta a_i$ once. This reference is held for the full external step. It is not recomputed as $v_{\text{stage}} + 6\Delta a_i$ at each stage. The controller below maps the reference to feasible rotor commands; actual acceleration depends on the evolving attitude and rotor limits.

F. Geometric feedback and feasible allocation

At a stage state, clip a held velocity reference to 20 m/s, then compute

$$\begin{aligned} a_d &= \text{sat}_6(2(v_d - v)), \\ F_d &= m(a_d + ge_3), \\ b_{3d} &= F_d/\|F_d\|, \\ b_{2d} &= (b_{3d} \times e_1)/\|b_{3d} \times e_1\|, \\ R_d &= [b_{2d} \times b_{3d}, b_{2d}, b_{3d}], \\ e_R &= \frac{1}{2}(R_d^T R - R^T R_d)^\vee, \\ M_d &= -8.81e_R - 2.54\omega + \omega \times J\omega, \\ f_d &= F_d^T R e_3. \end{aligned} \quad (117)$$

Fixed world-east heading avoids hidden yaw memory. The 6 m/s² bound is below g , so F_d and its heading construction are nonsingular. This uses attitude PD and gyro compensation, without the source paper's trajectory derivative or desired angular-rate feedforward.

To preserve feasible collective thrust, set $\bar{f} = \text{clip}(f_d, 0, 4f_{\max})/4$ and $d = A^{-1}(0, M_d)^T$. Define

$$\begin{aligned} \kappa_+ &= \min_{d_j > 0} (f_{\max} - \bar{f})/d_j, \\ \kappa_- &= \min_{d_j < 0} (-\bar{f})/d_j, \quad \kappa = \min(1, \kappa_+, \kappa_-). \end{aligned} \quad (118)$$

$$u_j = \text{clip}\left(\frac{\bar{f} + \kappa d_j}{f_{\max}}, 0, 1\right). \quad (119)$$

Empty minima are ignored. Uniform differential scaling sacrifices moment while retaining the clipped collective; independent rotor clipping is not assumed to preserve it.

G. Numerical and boundary rules

Use the synchronized 100-substep RK4 map (4) and clipped-path boundary rule on page 2. Hold the external command for 1 s; stage feedback is recomputed only for intent interfaces. Direct actuators remain constant.

XXVI. THEME 11 / QUADROTOR / ACCELERATION / SURVIVAL

Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 25, combined with the shared event rules on page 2, defines F_C . The configuration selects **Quadrotor / Acceleration** and **Survival**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Quadrotor event and scale instance

Use the constrained thirteen-value rigid state, $\lambda = .16$, $d_c = 3.2$ m and $(r_0, r_1) = (48, 96)$ m. Actor normalization uses $V_{\max} = 20$ m/s. Collision prediction uses synchronized, clipped RK4 path segments; rotor thrust follows the current body $+z$ direction. The acceleration action has three continuous coordinates or 27 unit primitives, scaled by 6 m/s^2 . The held velocity reference is constructed once from pre-step velocity. Stage feedback maps it to feasible rotor commands and preserves clipped collective thrust by uniform moment scaling. Other UAV attitudes/rates are not in entity rows, while the actor's own quaternion/rates are exposed. No motor lag, rotor drag or hidden controller integral is introduced.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 20$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\begin{aligned} \dim o_i &= 11(M-1) + 13, \\ \dim g &= 11M + 13N + 3. \end{aligned} \quad (120)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Survival reward and completion instance

Apply the Survival rule on page 3: asset HP decreases by raw L_k , and target loss has priority over Blue Attack elimination.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\text{term}} &= \mathbf{1}[z(s') \neq 0]. \end{aligned} \quad (121)$$

Blue receives $-R_t$. Red elimination alone does not end the world. A nonpositive-health actor retires after its lethal final transition, while the team return and successor value continue until global natural completion. Explicit optional reward weights replace this native scalar by the formulation on page 4; they do not add a second terminal bonus.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i \mid \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 25. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (122)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XXVII. THEME 12 / QUADROTOR / ACCELERATION / DAMAGE

Rigid-body thrust dynamics, reference feedback and feasible actuation.

A. Physical state for this interface

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state, health, role, side and replay records follow page 2.

B. Shared rigid-body transition

Use Newton–Euler (2) and RK4 (4) on page 2, with the wrench and controller specified here. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

C. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34$ kg, $J = \text{diag}(.0820, .0845, .1377)$ kg m², arm $l = .315$ m and yaw ratio $c_\tau = .008004$ m [2],

$$f_{\max} = 2.5mg/4, \quad f_j = f_{\max}u_j, \quad (123)$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \quad (124)$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \quad (125)$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z) . Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

D. Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400]$ m, $d_c = 3.2$ m, $(r_0, r_1) = (48, 96)$ m and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240]$ m or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352]$ m around a uniformly chosen asset, clipped to scene bounds; Blue is uniform over $[0, 400] \times [-400, 400]$ m. UAV altitude is uniform in $[32, 240]$ m.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6})$ m of any previously placed entity; later collisions are allowed. Only 3D is supported.

E. Acceleration intent

Each living agent chooses either $a_i \in [-1, 1]^3$ or an index into

$$\mathcal{V} = \{0\} \cup \{z/\|z\| : z \in \{-1, 0, 1\}^3 \setminus \{0\}\}, \quad (126)$$

giving 27 discrete primitives, zero first. The physical intent is $\alpha_i = 6a_i$ m/s². Continuous inputs have component bounds, so diagonal intents can have norm $\sqrt{3}6$; discrete nonzero intents have norm 6. 6 is a control scale, not a clamp on actual rigid-body acceleration. The external intent establishes $v_d = v(t) + 6\Delta a_i$ once. This reference is held for the full external step. It is not recomputed as $v_{\text{stage}} + 6\Delta a_i$ at each stage. The controller below maps the reference to feasible rotor commands; actual acceleration depends on the evolving attitude and rotor limits.

F. Geometric feedback and feasible allocation

At a stage state, clip a held velocity reference to 20 m/s, then compute

$$\begin{aligned} a_d &= \text{sat}_6(2(v_d - v)), \\ F_d &= m(a_d + ge_3), \\ b_{3d} &= F_d/\|F_d\|, \\ b_{2d} &= (b_{3d} \times e_1)/\|b_{3d} \times e_1\|, \\ R_d &= [b_{2d} \times b_{3d}, b_{2d}, b_{3d}], \\ e_R &= \frac{1}{2}(R_d^T R - R^T R_d)^\vee, \\ M_d &= -8.81e_R - 2.54\omega + \omega \times J\omega, \\ f_d &= F_d^T R e_3. \end{aligned} \quad (127)$$

Fixed world-east heading avoids hidden yaw memory. The 6 m/s² bound is below g , so F_d and its heading construction are nonsingular. This uses attitude PD and gyro compensation, without the source paper's trajectory derivative or desired angular-rate feedforward.

To preserve feasible collective thrust, set $\bar{f} = \text{clip}(f_d, 0, 4f_{\max})/4$ and $d = A^{-1}(0, M_d)^T$. Define

$$\begin{aligned} \kappa_+ &= \min_{d_j > 0} (f_{\max} - \bar{f})/d_j, \\ \kappa_- &= \min_{d_j < 0} (-\bar{f})/d_j, \quad \kappa = \min(1, \kappa_+, \kappa_-). \end{aligned} \quad (128)$$

$$u_j = \text{clip}\left(\frac{\bar{f} + \kappa d_j}{f_{\max}}, 0, 1\right). \quad (129)$$

Empty minima are ignored. Uniform differential scaling sacrifices moment while retaining the clipped collective; independent rotor clipping is not assumed to preserve it.

G. Numerical and boundary rules

Use the synchronized 100-substep RK4 map (4) and clipped-path boundary rule on page 2. Hold the external command for 1 s; stage feedback is recomputed only for intent interfaces. Direct actuators remain constant.

XXVIII. THEME 12 / QUADROTOR / ACCELERATION / DAMAGE

Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 27, combined with the shared event rules on page 2, defines F_C . The configuration selects **Quadrotor / Acceleration** and **Damage**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Quadrotor event and scale instance

Use the constrained thirteen-value rigid state, $\lambda = .16$, $d_c = 3.2$ m and $(r_0, r_1) = (48, 96)$ m. Actor normalization uses $V_{\max} = 20$ m/s. Collision prediction uses synchronized, clipped RK4 path segments; rotor thrust follows the current body $+z$ direction. The acceleration action has three continuous coordinates or 27 unit primitives, scaled by 6 m/s^2 . The held velocity reference is constructed once from pre-step velocity. Stage feedback maps it to feasible rotor commands and preserves clipped collective thrust by uniform moment scaling. Other UAV attitudes/rates are not in entity rows, while the actor's own quaternion/rates are exposed. No motor lag, rotor drag or hidden controller integral is introduced.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 20$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\begin{aligned} \dim o_i &= 11(M-1) + 13, \\ \dim g &= 11M + 13N + 3. \end{aligned} \quad (130)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Damage reward and completion instance

Apply the Damage rule on page 3: finite asset HP stays at reset health; raw simultaneous damage is counted before clipping or overkill normalization.

$$\begin{aligned} h'_k &= h_k^0, & D'_k &= D_k + L_k, \\ R_t &= -\sum_k L_k, & d_t^{\text{term}} &= \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0]. \end{aligned} \quad (131)$$

Blue receives $-R_t$. Every team slot receives the full shared reward, including dead Red slots and later damage after Red extinction; summing duplicate teammate rewards changes the objective. Dead slots keep zero observations and ignored actions until global completion/truncation. At $\gamma = 1$, the Red return is $-\sum_k D_k$, with no Survival terminal bonus or optional Survival weights.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i | \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 27. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (132)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XXIX. THEME 13 / QUADROTOR / ACTUATOR / SURVIVAL
Rigid-body thrust dynamics, reference feedback and feasible actuation.

A. Physical state for this interface

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state, health, role, side and replay records follow page 2.

B. Shared rigid-body transition

Use Newton–Euler (2) and RK4 (4) on page 2, with the wrench and controller specified here. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

C. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34$ kg, $J = \text{diag}(.0820, .0845, .1377)$ kg m², arm $l = .315$ m and yaw ratio $c_\tau = .008004$ m [2],

$$f_{\max} = 2.5mg/4, \quad f_j = f_{\max}u_j, \quad (133)$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \quad (134)$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \quad (135)$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z) . Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

D. Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400]$ m, $d_c = 3.2$ m, $(r_0, r_1) = (48, 96)$ m and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240]$ m or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352]$ m around a uniformly chosen asset, clipped to scene bounds; Blue is uniform over $[0, 400] \times [-400, 400]$ m. UAV altitude is uniform in $[32, 240]$ m.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6})$ m of any previously placed entity; later collisions are allowed. Only 3D is supported.

E. Direct rotor-thrust action

Each living agent selects $a_i \in [0, 1]^4$, ordered front, right, rear, left, and $u = a_i$ is held throughout the external step. The wrench map enters Newton-Euler dynamics directly, without automatic attitude or position stabilization. Zero thrust is free fall, not hover; .4 on every rotor is ideal level hover.

F. Numerical and boundary rules

Use the synchronized 100-substep RK4 map (4) and clipped-path boundary rule on page 2. Hold the external command for 1 s; stage feedback is recomputed only for intent interfaces. Direct actuators remain constant.

G. Decision and task interpretation

The agent must learn a rotor-level feedback policy from its local information. Identical rotor actions can generate different world accelerations because thrust follows the current body $+z$ axis. Attitude and angular velocity are therefore indispensable components of the physical state and are explicitly supplied in the actor's self block.

The default world is $[-400, 400]^2 \times [0, 400]$ m. Protected assets have positive finite health in both tasks; only the Survival task decrements asset health. A saturated or destabilizing action is propagated by the rigid-body equations, rather than repaired by a reference controller. Low altitude does not itself cause a crash: scene boundaries project position and remove outward velocity.

XXX. THEME 13 / QUADROTOR / ACTUATOR / SURVIVAL
Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 29, combined with the shared event rules on page 2, defines F_C . The configuration selects **Quadrotor / Actuator** and **Survival**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Quadrotor event and scale instance

Use the constrained thirteen-value rigid state, $\lambda = .16$, $d_c = 3.2$ m and $(r_0, r_1) = (48, 96)$ m. Actor normalization uses $V_{\max} = 20$ m/s. Collision prediction uses synchronized, clipped RK4 path segments; rotor thrust follows the current body $+z$ direction. The direct actuator action has four continuous thrust fractions in $[0, 1]$, ordered front, right, rear, left. The fractions are held for the full step. Zero action is free fall, while .4 on all rotors is ideal level hover; a local learned policy supplies stabilization. Other UAV attitudes/rates are not in entity rows, while the actor's own quaternion/rates are exposed. No motor lag, rotor drag or hidden controller integral is introduced.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M - 1) \times 11$ entity table ($V_{\max} = 20$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Rotor controls do not add an observation block. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\begin{aligned} \dim o_i &= 11(M - 1) + 13, \\ \dim g &= 11M + 13N + 3. \end{aligned} \quad (136)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = 1[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Survival reward and completion instance

Apply the Survival rule on page 3: asset HP decreases by raw L_k , and target loss has priority over Blue Attack elimination.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\text{term}} &= 1[z(s') \neq 0]. \end{aligned} \quad (137)$$

Blue receives $-R_t$. Red elimination alone does not end the world. A nonpositive-health actor retires after its lethal final transition, while the team return and successor value continue until global natural completion. Explicit optional reward weights replace this native scalar by the formulation on page 4; they do not add a second terminal bonus.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i \mid \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 29. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (138)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XXXI. THEME 14 / QUADROTOR / ACTUATOR / DAMAGE

Rigid-body thrust dynamics, reference feedback and feasible actuation.

A. Physical state for this interface

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state, health, role, side and replay records follow page 2.

B. Shared rigid-body transition

Use Newton–Euler (2) and RK4 (4) on page 2, with the wrench and controller specified here. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

C. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34$ kg, $J = \text{diag}(.0820, .0845, .1377)$ kg m², arm $l = .315$ m and yaw ratio $c_\tau = .008004$ m [2],

$$f_{\max} = 2.5mg/4, \quad f_j = f_{\max}u_j, \quad (139)$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \quad (140)$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \quad (141)$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z) . Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

D. Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400]$ m, $d_c = 3.2$ m, $(r_0, r_1) = (48, 96)$ m and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240]$ m or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352]$ m around a uniformly chosen asset, clipped to scene bounds; Blue is uniform over $[0, 400] \times [-400, 400]$ m. UAV altitude is uniform in $[32, 240]$ m.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6})$ m of any previously placed entity; later collisions are allowed. Only 3D is supported.

E. Direct rotor-thrust action

Each living agent selects $a_i \in [0, 1]^4$, ordered front, right, rear, left, and $u = a_i$ is held throughout the external step. The wrench map enters Newton-Euler dynamics directly, without automatic attitude or position stabilization. Zero thrust is free fall, not hover; .4 on every rotor is ideal level hover.

F. Numerical and boundary rules

Use the synchronized 100-substep RK4 map (4) and clipped-path boundary rule on page 2. Hold the external command for 1 s; stage feedback is recomputed only for intent interfaces. Direct actuators remain constant.

G. Decision and task interpretation

The agent must learn a rotor-level feedback policy from its local information. Identical rotor actions can generate different world accelerations because thrust follows the current body +z axis. Attitude and angular velocity are therefore indispensable components of the physical state and are explicitly supplied in the actor's self block.

The default world is $[-400, 400]^2 \times [0, 400]$ m. Protected assets have positive finite health in both tasks; only the Survival task decrements asset health. A saturated or destabilizing action is propagated by the rigid-body equations, rather than repaired by a reference controller. Low altitude does not itself cause a crash: scene boundaries project position and remove outward velocity.

XXXII. THEME 14 / QUADROTOR / ACTUATOR / DAMAGE
Defender task instance, information dimensions and learning
contract.

A. Controlled process for this theme

The physical map on page 31, combined with the shared event rules on page 2, defines F_C . The configuration selects **Quadrotor / Actuator** and **Damage**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Quadrotor event and scale instance

Use the constrained thirteen-value rigid state, $\lambda = .16$, $d_c = 3.2$ m and $(r_0, r_1) = (48, 96)$ m. Actor normalization uses $V_{\max} = 20$ m/s. Collision prediction uses synchronized, clipped RK4 path segments; rotor thrust follows the current body $+z$ direction. The direct actuator action has four continuous thrust fractions in $[0, 1]$, ordered front, right, rear, left. The fractions are held for the full step. Zero action is free fall, while .4 on all rotors is ideal level hover; a local learned policy supplies stabilization. Other UAV attitudes/rates are not in entity rows, while the actor's own quaternion/rates are exposed. No motor lag, rotor drag or hidden controller integral is introduced.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 20$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\begin{aligned} \dim o_i &= 11(M-1) + 13, \\ \dim g &= 11M + 13N + 3. \end{aligned} \quad (142)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = 1[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Damage reward and completion instance

Apply the Damage rule on page 3: finite asset HP stays at reset health; raw simultaneous damage is counted before clipping or overkill normalization.

$$\begin{aligned} h'_k &= h_k^0, & D'_k &= D_k + L_k, \\ R_t &= -\sum_k L_k, & d_t^{\text{term}} &= 1[\forall b \in \mathcal{B}_A : h'_b \leq 0]. \end{aligned} \quad (143)$$

Blue receives $-R_t$. Every team slot receives the full shared reward, including dead Red slots and later damage after Red extinction; summing duplicate teammate rewards changes the objective. Dead slots keep zero observations and ignored actions until global completion/truncation. At $\gamma = 1$, the Red return is $-\sum_k D_k$, with no Survival terminal bonus or optional Survival weights.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i | \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 31. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (144)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XXXIII. THEME 15 / QUADROTOR / POSITION / SURVIVAL

Rigid-body thrust dynamics, reference feedback and feasible actuation.

A. Physical state for this interface

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state, health, role, side and replay records follow page 2.

B. Shared rigid-body transition

Use Newton–Euler (2) and RK4 (4) on page 2, with the wrench and controller specified here. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

C. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34$ kg, $J = \text{diag}(.0820, .0845, .1377)$ kg m², arm $l = .315$ m and yaw ratio $c_\tau = .008004$ m [2],

$$f_{\max} = 2.5mg/4, \quad f_j = f_{\max}u_j, \quad (145)$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \quad (146)$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \quad (147)$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z) . Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

D. Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400]$ m, $d_c = 3.2$ m, $(r_0, r_1) = (48, 96)$ m and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240]$ m or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352]$ m around a uniformly chosen asset, clipped to scene bounds; Blue is uniform over $[0, 400] \times [-400, 400]$ m. UAV altitude is uniform in $[32, 240]$ m.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6})$ m of any previously placed entity; later collisions are allowed. Only 3D is supported.

E. Absolute position intent

Each living agent selects $p_d \in \mathcal{W}$ in metres. At every RK4 stage compute

$$v_d = \text{sat}_{12}(.5(p_d - p)), \quad \text{sat}_L(z) = z \min(1, L/\|z\|). \quad (148)$$

For $z = 0$, define $\text{sat}_L(0) = 0$. The position reference is held over the external step; feedback updates using stage position and velocity. Unlike the fixed wing and minimum-speed particle, this ideal quadrotor can command zero velocity near its target. Reaching the point is a feedback outcome, not an instantaneous assignment.

F. Geometric feedback and feasible allocation

At a stage state, clip a held velocity reference to 20 m/s, then compute

$$\begin{aligned} a_d &= \text{sat}_6(2(v_d - v)), \\ F_d &= m(a_d + ge_3), \\ b_{3d} &= F_d/\|F_d\|, \\ b_{2d} &= (b_{3d} \times e_1)/\|b_{3d} \times e_1\|, \\ R_d &= [b_{2d} \times b_{3d}, b_{2d}, b_{3d}], \\ e_R &= \frac{1}{2}(R_d^T R - R^T R_d)^\vee, \\ M_d &= -8.81e_R - 2.54\omega + \omega \times J\omega, \\ f_d &= F_d^T R e_3. \end{aligned} \quad (149)$$

Fixed world-east heading avoids hidden yaw memory. The 6 m/s² bound is below g , so F_d and its heading construction are nonsingular. This uses attitude PD and gyro compensation, without the source paper's trajectory derivative or desired angular-rate feedforward.

To preserve feasible collective thrust, set $\bar{f} = \text{clip}(f_d, 0, 4f_{\max})/4$ and $d = A^{-1}(0, M_d)^T$. Define

$$\begin{aligned} \kappa_+ &= \min_{d_j > 0} (f_{\max} - \bar{f})/d_j, \\ \kappa_- &= \min_{d_j < 0} (-\bar{f})/d_j, \quad \kappa = \min(1, \kappa_+, \kappa_-). \end{aligned} \quad (150)$$

$$u_j = \text{clip}\left(\frac{\bar{f} + \kappa d_j}{f_{\max}}, 0, 1\right). \quad (151)$$

Empty minima are ignored. Uniform differential scaling sacrifices moment while retaining the clipped collective; independent rotor clipping is not assumed to preserve it.

G. Numerical and boundary rules

Use the synchronized 100-substep RK4 map (4) and clipped-path boundary rule on page 2. Hold the external command for 1 s; stage feedback is recomputed only for intent interfaces. Direct actuators remain constant.

XXXIV. THEME 15 / QUADROTOR / POSITION / SURVIVAL
Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 33, combined with the shared event rules on page 2, defines F_C . The configuration selects **Quadrotor / Position** and **Survival**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Quadrotor event and scale instance

Use the constrained thirteen-value rigid state, $\lambda = .16$, $d_c = 3.2$ m and $(r_0, r_1) = (48, 96)$ m. Actor normalization uses $V_{\max} = 20$ m/s. Collision prediction uses synchronized, clipped RK4 path segments; rotor thrust follows the current body $+z$ direction. The position action has three continuous coordinates in $\mathcal{W}$. Stage feedback uses a 12 m/s norm-limited position-derived velocity reference, then the common geometric controller and feasible rotor allocation. Zero reference velocity near a point is allowed. Other UAV attitudes/rates are not in entity rows, while the actor's own quaternion/rates are exposed. No motor lag, rotor drag or hidden controller integral is introduced.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 20$) plus unnormalized $[p_i, v_i, q_i, \omega_i] \in \mathbb{R}^{13}$. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\begin{aligned} \dim o_i &= 11(M-1) + 13, \\ \dim g &= 11M + 13N + 3. \end{aligned} \quad (152)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = 1[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Survival reward and completion instance

Apply the Survival rule on page 3: asset HP decreases by raw L_k , and target loss has priority over Blue Attack elimination.

$$\begin{aligned} h'_k &= \max(0, h_k - L_k), & D'_k &= D_k + L_k, \\ R_t &= 10z(s'), & d_t^{\text{term}} &= 1[z(s') \neq 0]. \end{aligned} \quad (153)$$

Blue receives $-R_t$. Red elimination alone does not end the world. A nonpositive-health actor retires after its lethal final transition, while the team return and successor value continue until global natural completion. Explicit optional reward weights replace this native scalar by the formulation on page 4; they do not add a second terminal bonus.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i | \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 33. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (154)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.

XXXV. THEME 16 / QUADROTOR / POSITION / DAMAGE

Rigid-body thrust dynamics, reference feedback and feasible actuation.

A. Physical state for this interface

$x_i = [p_i^T, v_i^T, q_i^T, \omega_i^T]^T \in \mathbb{R}^{13}$ with $\|q_i\| = 1$. The complete state, health, role, side and replay records follow page 2.

B. Shared rigid-body transition

Use Newton–Euler (2) and RK4 (4) on page 2, with the wrench and controller specified here. ENU/FLU axes and normalized scalar-first body-to-world quaternions apply.

C. Four-rotor wrench and parameters

The rotor order is front, right, rear, left, and normalized thrust fractions are $u_j \in [0, 1]$. With $m = 4.34$ kg, $J = \text{diag}(.0820, .0845, .1377)$ kg m², arm $l = .315$ m and yaw ratio $c_\tau = .008004$ m [2],

$$f_{\max} = 2.5mg/4, \quad f_j = f_{\max}u_j, \quad (155)$$

$$\begin{bmatrix} f \\ M_x \\ M_y \\ M_z \end{bmatrix} = A \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \quad (156)$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -l & 0 & l \\ -l & 0 & l & 0 \\ c_\tau & -c_\tau & c_\tau & -c_\tau \end{bmatrix}. \quad (157)$$

The rigid-body force is $F_B = (0, 0, f)$ and moment is (M_x, M_y, M_z) . Ideal level hover has $u_j = .4$. The maximum thrust-to-weight ratio 2.5 is a simulation assumption. No aerodynamic/rotor drag, gyroscopic rotor torque or motor time constant is included [3].

D. Episode geometry and reset law

Default $\lambda = .16$, $\mathcal{W} = [-400, 400]^2 \times [0, 400]$ m, $d_c = 3.2$ m, $(r_0, r_1) = (48, 96)$ m and $r_f = r_0$ (configurable $0 < r_f \leq r_0$). Assets are independently uniform in $[-368, -304] \times [-192, 192] \times [80, 240]$ m or use a fixed layout. Red samples an area-uniform XY annulus $[128, 352]$ m around a uniformly chosen asset, clipped to scene bounds; Blue is uniform over $[0, 400] \times [-400, 400]$ m. UAV altitude is uniform in $[32, 240]$ m.

Reset UAVs at $v = 0$, $q = (1, 0, 0, 0)$, $\omega = 0$. Reject UAV openings within $\max(4.8, d_c + 10^{-6})$ m of any previously placed entity; later collisions are allowed. Only 3D is supported.

E. Absolute position intent

Each living agent selects $p_d \in \mathcal{W}$ in metres. At every RK4 stage compute

$$v_d = \text{sat}_{12}(.5(p_d - p)), \quad \text{sat}_L(z) = z \min(1, L/\|z\|). \quad (158)$$

For $z = 0$, define $\text{sat}_L(0) = 0$. The position reference is held over the external step; feedback updates using stage position and velocity. Unlike the fixed wing and minimum-speed particle, this ideal quadrotor can command zero velocity near its target. Reaching the point is a feedback outcome, not an instantaneous assignment.

F. Geometric feedback and feasible allocation

At a stage state, clip a held velocity reference to 20 m/s, then compute

$$\begin{aligned} a_d &= \text{sat}_6(2(v_d - v)), \\ F_d &= m(a_d + ge_3), \\ b_{3d} &= F_d/\|F_d\|, \\ b_{2d} &= (b_{3d} \times e_1)/\|b_{3d} \times e_1\|, \\ R_d &= [b_{2d} \times b_{3d}, b_{2d}, b_{3d}], \\ e_R &= \frac{1}{2}(R_d^T R - R^T R_d)^\vee, \\ M_d &= -8.81e_R - 2.54\omega + \omega \times J\omega, \\ f_d &= F_d^T R e_3. \end{aligned} \quad (159)$$

Fixed world-east heading avoids hidden yaw memory. The 6 m/s² bound is below g , so F_d and its heading construction are nonsingular. This uses attitude PD and gyro compensation, without the source paper's trajectory derivative or desired angular-rate feedforward.

To preserve feasible collective thrust, set $\bar{f} = \text{clip}(f_d, 0, 4f_{\max})/4$ and $d = A^{-1}(0, M_d)^T$. Define

$$\begin{aligned} \kappa_+ &= \min_{d_j > 0} (f_{\max} - \bar{f})/d_j, \\ \kappa_- &= \min_{d_j < 0} (-\bar{f})/d_j, \quad \kappa = \min(1, \kappa_+, \kappa_-). \end{aligned} \quad (160)$$

$$u_j = \text{clip}\left(\frac{\bar{f} + \kappa d_j}{f_{\max}}, 0, 1\right). \quad (161)$$

Empty minima are ignored. Uniform differential scaling sacrifices moment while retaining the clipped collective; independent rotor clipping is not assumed to preserve it.

G. Numerical and boundary rules

Use the synchronized 100-substep RK4 map (4) and clipped-path boundary rule on page 2. Hold the external command for 1 s; stage feedback is recomputed only for intent interfaces. Direct actuators remain constant.

XXXVI. THEME 16 / QUADROTOR / POSITION / DAMAGE

Defender task instance, information dimensions and learning contract.

A. Controlled process for this theme

The physical map on page 35, combined with the shared event rules on page 2, defines F_C . The configuration selects **Quadrotor / Position** and **Damage**; it leaves episode-fixed Attack, Scout and Disturb roles intact. Red controls navigation; each living Blue agent receives its own navigation action from the specified external Blue policy. Automatic firing is the same separate rule for both teams.

B. Quadrotor event and scale instance

Use the constrained thirteen-value rigid state, $\lambda = .16$, $d_c = 3.2$ m and $(r_0, r_1) = (48, 96)$ m. Actor normalization uses $V_{\max} = 20$ m/s. Collision prediction uses synchronized, clipped RK4 path segments; rotor thrust follows the current body $+z$ direction. The position action has three continuous coordinates in $\mathcal{W}$. Stage feedback uses a 12 m/s norm-limited position-derived velocity reference, then the common geometric controller and feasible rotor allocation. Zero reference velocity near a point is allowed. Other UAV attitudes/rates are not in entity rows, while the actor's own quaternion/rates are exposed. No motor lag, rotor drag or hidden controller integral is introduced.

C. Transition and event checklist

First predict all living paths and resolve UAV collisions in fixed pair order. At step-start positions, compute automatic Attack fire and Disturb decisions from the common post-collision snapshot. Add all interaction losses before clipping health, then perform the controlled motion of collision survivors, including agents killed by combat. Update the selected task, increment the step, and pack successor observations. Inactive velocities are zeroed at the next update.

The default firing radius is $r_f = r_0$; configuration can choose $0 < r_f \leq r_0$. Fire may affect multiple enemies. There is no invented visibility mask: every living entity has an actor row irrespective of role sectors or firing range.

D. Exact local and critic inputs

The actor receives the $(M-1) \times 11$ entity table ($V_{\max} = 20$) plus unnormalized $[p_i, v_i, q_i, \omega_i]$. Both blocks are zero for an already-dead Damage observer. Entity rows are the normalized eleven-value encoding on page 3, with fixed roster order, dead rows zero and no own-health entry in the self block.

$$\begin{aligned} \dim o_i &= 11(M-1) + 13, \\ \dim g &= 11M + 13N + 3. \end{aligned} \quad (162)$$

The entity validity mask indicates living entries. Action participation is $\ell_{i,t} = \mathbf{1}[h_{i,t} > 0]$ before the transition. Role labels, policy memory and absolute time are not inferable merely from side flags. A critic receives the privileged packing only by an explicit CTDE choice; exact value conditioning additionally requires relevant task state and policy memory/history as on page 4.

E. Damage reward and completion instance

Apply the Damage rule on page 3: finite asset HP stays at reset health; raw simultaneous damage is counted before clipping or overkill normalization.

$$\begin{aligned} h'_k &= h_k^0, & D'_k &= D_k + L_k, \\ R_t &= -\sum_k L_k, & d_t^{\text{term}} &= \mathbf{1}[\forall b \in \mathcal{B}_A : h'_b \leq 0]. \end{aligned} \quad (163)$$

Blue receives $-R_t$. Every team slot receives the full shared reward, including dead Red slots and later damage after Red extinction; summing duplicate teammate rewards changes the objective. Dead slots keep zero observations and ignored actions until global completion/truncation. At $\gamma = 1$, the Red return is $-\sum_k D_k$, with no Survival terminal bonus or optional Survival weights.

F. Defender learning instance

Freeze a specified external π_B and integrate its navigation actions and memory into T^{π_B} on page 3. Choose decentralized Red actors $\pi_i(a_i | \tau_i)$ using local histories and the above interface. Then optimize the continuing objective on page 4 with this task's R_t and reset law from page 35. Jointly adapting both teams instead defines the corresponding POSG.

$$\begin{aligned} b_t &= 1 - d_t^{\text{term}}, \\ y_t &= R_t + \gamma b_t Q^-(\xi_{t+1}, a'_R). \end{aligned} \quad (164)$$

The caller's horizon H truncates a sample, including after individual death; it does not award native victory. Retain successor value at truncation. Natural completion has priority when both occur. Target actions must use the successor policy memories; ξ includes relevant Red recurrent memory and frozen Blue memory. Actor loss uses pre-step participation, so a lethal action remains trainable.

This is a mathematical task and information contract. It specifies neither optimizer nor a reported training result. Grouping on page 4 is an optional acceleration execution extension, rather than another native theme.